



Temporal Equivalence Principle: A Blind-Prediction Residual Test in Multiply-Imaged Supernovae

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Abstract

The Temporal Equivalence Principle (TEP) predicts that light rays traversing different gravitational potential depths experience differential temporal scaling ($\alpha \neq 0$). This work applies a geometric blind-prediction residual test to SN Refsdal, the only multiply-imaged supernova with five resolved images and precision-measured time delays. The key observational lever is a comparison of seven blind pre-reappearance GR predictions of $\Delta t_{\text{SX},\text{S1}}$ against the later Kelly et al. (2023) measurement — two completely disjoint datasets.

The pre-specified primary non-parametric directional test is a Wilcoxon signed-rank test on the six non-zero residuals among the seven blind models, all of which are positive ($p = 0.016$, approximately 2.2σ), matching the log-magnification proxy prediction for a negative temporal-shear coupling. The supplementary all-eight-model Wilcoxon (including one post-blind precision update) gives $p = 0.0078$ (2.4σ). Additional strands are directionally consistent: a 46% reduction in weighted RMS after proxy-model correction, and a binomial sign test of 6/7 positive among blind models ($p = 0.0625$). Hierarchical Bayesian model comparison remains inconclusive ($\text{BF} = 1.06$ fixed- α baseline / 0.997 h0pe-informed, $\text{BF} = 0.615$ free- α baseline / 0.492 h0pe-informed with a GR-centred prior), indicating that present lens-model uncertainties dominate formal model-selection metrics. The measured coupling $\alpha_{\text{lens}} = -0.055 \pm 0.044$ is calibrated from the same SN Refsdal data, so the magnitude agreement with the proxy-model prediction is definitional rather than an independent confirmation; the probative content lies in the sign consistency across modelling groups that use independent codes but share the same lens and image constraints. In the absence of a solved TEP lensing transfer function for $\kappa(\theta)$ or $\psi(\theta)$, this test should be read as a falsifiable phenomenological screen, not a fundamental coupling measurement. Applying this coupling shifts SN Refsdal's inferred H_0 from 66.6 to 69.3 $\text{km s}^{-1} \text{Mpc}^{-1}$ as an internal consistency check; corrections to SN Encore and SN H0pe are negligible ($\sim 0.1 \text{ km s}^{-1} \text{Mpc}^{-1}$) due to their smaller magnification contrasts.

1. Introduction

The Temporal Equivalence Principle (TEP) posits that the effective rate of temporal propagation scales with the depth of the local gravitational potential. This hypothesis emerges from a scalar-field extension to general relativity in which the temporal metric component acquires a potential-dependent coupling, analogous to how the spatial metric component encodes gravitational redshift. In such a framework, photons traversing regions of different gravitational potential depth experience differential temporal scaling—a phenomenon termed "temporal shear." This hypothesis has previously been explored in stellar evolution anomalies, Cepheid period–luminosity residuals, and galaxy-scale redshift correlations (see §References). The strong gravitational lensing regime provides a qualitatively different and entirely independent test. Here, a single photon traverses multiple geometric paths through a cluster or galaxy potential, accumulating a potential-dependent temporal shear along each sightline. The resulting pairwise time delays between images carry a direct imprint of the differential shear—independently of any cosmological model.

1.1 The Blind-Prediction Residual Test

For a source producing three or more resolved images, the pairwise time delays obey a strict algebraic closure identity under any theory in which each image has a single, globally assignable arrival time: any three delays in a closed loop sum to identically zero. This identity holds for GR and for the TEP ansatz alike, because TEP modifies the effective transit time along each path but still assigns a unique arrival time to each image. The genuine TEP observable is not a closure violation but a *blind-prediction residual*: the discrepancy between the observed delay and the delay predicted by a GR lens model that assumes standard time propagation. In the fundamental theory, light along path i through a region of projected convergence κ_i acquires a temporal shear that scales with the local gravitational potential depth. In the absence of a solved TEP lensing transfer function for $\kappa(\boldsymbol{\theta})$ or $\psi(\boldsymbol{\theta})$, we adopt a first-order **log-magnification phenomenological proxy model**:

$$\Gamma_t(i) = 1 + \alpha \log_{10}(\mu_i),$$

where μ_i is the total magnification at image i and α is the proxy-model coupling. The predicted GR-vs-TEP discrepancy for loop (i, j, k) ,

$$\mathcal{R}_{\text{TEP/GR}}(i, j, k) = (\Gamma_i - 1)\Delta t_{ij} + (\Gamma_j - 1)\Delta t_{jk} + (\Gamma_k - 1)\Delta t_{ki} \quad (1)$$

is non-zero if and only if the images traverse regions of different potential depth. This predicted discrepancy is immune to the Mass Sheet Degeneracy: a uniform convergence sheet rescales all delays by the same factor, leaving the algebraic loop sum unchanged at zero under both GR and TEP. The blind-prediction residual test is therefore a clean probe of potential-dependent temporal propagation.

1.2 SN Refsdal: The Ideal System

SN Refsdal (MACS J1149.6+2223, Kelly et al. 2015) is the only known multiply-imaged supernova with *five* resolved images and precision-measured relative time delays. The first four images (S1–S4) form an Einstein cross around a cluster member galaxy; the fifth image SX appeared ~ 8 arcsec away at a separate arc position approximately 376 days after S1, predicted by cluster lens models before it was observed. Kelly et al. (2023, ApJ 948, 93) published the definitive time delay measurements from the combined light curve analysis, including the SX–S1 delay measured to 1.5% precision—the most precise lensed supernova delay to date.

The geometry of SN Refsdal is ideal for the blind-prediction residual test. The four Einstein-cross images (S1–S4) sample the deep potential of the cluster member galaxy halo, while SX samples the outer cluster arc at much lower magnification. This contrast in potential depth—combined with the long 376-day SX baseline—amplifies the expected TEP predicted discrepancy by a factor of ~ 18 relative to the inner cross loops alone.

The test is structurally a **single-contrast measurement**: the S4–SX magnification difference provides the sole predictive leverage. The inner Einstein cross is a null region where the proxy is non-predictive because shear degeneracy renders the magnification ordering uninformative about convergence (Step 33); the five-image rank-order agreement between flux-proxy μ_{norm} and inferred κ_{norm} drops to $P \approx 3\%$. The sole testable contrast is the long-baseline S4–SX pair.

1.3 TDCOSMO Quad Lenses: Sub-Noise Supplementary Check

As a supplementary structural check, proxy-model predicted delay shifts are computed for eight quad-lens quasar systems from the expanded TDCOSMO-2025 dataset plus the newly observed SN Encore. These systems have shorter delay baselines ($\lesssim 160$ days) and moderate magnification contrasts, yielding predicted proxy shifts of 0.03–4.7 days at the measured coupling $\alpha_{\text{lens}} \approx -0.055$. At

this coupling, 16 of 18 image pairs exhibit a predicted shift exceeding the 1σ measurement uncertainty. The Spearman rank correlation between logarithmic flux ratio and predicted shift is tautological ($\rho < 0$ by construction because $\alpha_{\text{lens}} < 0$; computed $\rho = -0.733$) and therefore carries no independent information; the physically meaningful quantity is the predicted shift magnitude versus published delay uncertainties. Critically, these systems do not permit a full geometric blind-prediction residual test because all independent pairwise delays are referenced to the same reference image, making any loop sum arithmetically zero by construction.

1.4 The Core Evidence Approach: Observed vs. Blind-Predicted

The strongest observational lever available with current data is a feature of SN Refsdal's discovery history that has not previously been exploited as a TEP test: seven independent GR lens modelling teams published blind predictions for the $\Delta t_{\text{SX},\text{S1}}$ delay before SX reappeared in December 2015 (compiled in Treu et al. 2016, ApJ 817, 60). Kelly et al. (2023) then independently measured the delay from SN light-curve fitting—using completely disjoint data. The comparison of these two independent determinations constitutes a genuine non-trivial residual test: the residual $\mathcal{R}_{\text{obs}} = \Delta t_{\text{obs}} - \langle \Delta t_{\text{model}} \rangle$ is not constrained to be zero. Six of the seven blind models give positive residuals, while the seventh (Diego, WSLAP+) lies exactly at the observed value ($\delta = 0$, consistent with GR). The sign and magnitude are consistent with the proxy-model prediction. In the absence of a solved TEP lensing transfer function for $\kappa(\boldsymbol{\theta})$ or $\psi(\boldsymbol{\theta})$, this test should be read as a falsifiable phenomenological screen, not a fundamental coupling measurement. This is the primary evidence result of this paper.

1.5 SN 2025wny: Forward Prediction

The discovery of SN 2025wny ($z_s = 2.011$, Johansson et al. 2025, ApJ 995, L17)—the first resolved quadruply-imaged superluminous supernova—provides a falsifiable forward prediction target. With magnifications estimated at $\mu \sim 20\text{--}50$ and four images in an Einstein cross geometry, the system is expected to exhibit proxy-model predicted discrepancies of order 1–15 days once time delays are measured.

2. Methodology

2.1 Standard GR Time Delay

In General Relativity, the observed time delay between images i and j of a lensed source is given by the Fermat potential difference:

$$\Delta t_{ij} = \frac{D_{\Delta t}}{c} \left[\frac{1}{2}(\boldsymbol{\theta}_i - \boldsymbol{\beta})^2 - \psi(\boldsymbol{\theta}_i) - \frac{1}{2}(\boldsymbol{\theta}_j - \boldsymbol{\beta})^2 + \psi(\boldsymbol{\theta}_j) \right] \quad (2)$$

where $D_{\Delta t} = (1 + z_l)D_l D_s / D_{ls}$ is the time-delay distance, $\boldsymbol{\beta}$ is the unlensed source position, $\boldsymbol{\theta}_i$ are the image positions, and $\psi(\boldsymbol{\theta})$ is the projected lens potential. Each image has a unique absolute arrival time t_i ; any pairwise delay is just $\Delta t_{ij} = t_j - t_i$.

2.2 The GR Algebraic Loop Identity

For any three images (i, j, k) from the same source, the oriented sum of pairwise delays must vanish identically under any theory that assigns a single, globally well-defined arrival time to each image:

$$\mathcal{L}(i, j, k) \equiv \Delta t_{ij} + \Delta t_{jk} + \Delta t_{ki} = (t_j - t_i) + (t_k - t_j) + (t_i - t_k) = 0 \quad (3)$$

This is a purely algebraic identity, independent of the lens model, cosmology, or the Mass Sheet Degeneracy. It holds for GR and for the TEP ansatz alike, because TEP modifies the transit time along each path but does not introduce path-dependent holonomy that would break global time assignment. It holds for *any* combination of three images from five, giving $\binom{5}{3} = 10$ possible loops for SN Refsdal, of which five are physically informative and used in this analysis.

2.3 TEP Temporal Shear: The Log-Magnification Phenomenological Proxy Model

In the absence of a solved TEP lensing transfer function for $\kappa(\boldsymbol{\theta})$ or $\psi(\boldsymbol{\theta})$, we adopt a first-order **log-magnification phenomenological proxy model**. The TEP coupling is fundamentally to the projected convergence $\kappa(\boldsymbol{\theta})$ at each image position. The effective transit time of light along path i under a convergence-proxy model is:

$$\Gamma_t^{(\kappa)}(i) = 1 + \alpha_\kappa \log_{10}(\kappa_{\text{norm}}(i)) \quad (4)$$

where $\kappa_{\text{norm}}(i) = \kappa(i)/\bar{\kappa}$. Because direct convergence values from high-resolution mass models are not yet available for every image, the operational analysis uses the published total flux ratios F_i/F_{ref} as a proxy for magnification. This introduces the **proxy model**:

$$\Delta t_i^{\text{obs}} = \Delta t_i^{\text{GR}} \cdot \Gamma_t^{\text{proxy}}(i), \quad \Gamma_t^{\text{proxy}}(i) = 1 + \alpha \log_{10}(\mu_{\text{norm}}(i)) \quad (5)$$

where $\mu_{\text{norm}}(i) = \mu(i)/\bar{\mu}$ is the magnification at image i normalised to the mean across all images, and α is the proxy-model temporal-shear coupling parameter. The lensing-sector effective coupling is determined empirically from the SN Refsdal data: $\alpha_{\text{lens}} = -0.055 \pm 0.044$ (§3.5). Because $\alpha < 0$, images with below-average magnification ($\mu_{\text{norm}} < 1$, $\log_{10}(\mu_{\text{norm}}) < 0$) experience a temporal expansion ($\Gamma_t > 1$) and arrive *later* than GR predicts, while highly magnified images ($\mu_{\text{norm}} > 1$) arrive *earlier* ($\Gamma_t < 1$).

The systematic error introduced by the proxy is the difference between the convergence-proxy and flux-proxy temporal shear factors:

$$\delta\Gamma_t(i) \equiv \Gamma_t^{\text{proxy}}(i) - \Gamma_t^{(\kappa)}(i) = \alpha [\log_{10}(\mu_{\text{norm}}(i)) - \log_{10}(\kappa_{\text{norm}}(i))] \quad (6)$$

This systematic vanishes only when $\mu_{\text{norm}}(i) = \kappa_{\text{norm}}(i)$ for all images, which requires the shear γ to be negligible or identical across images. In cluster lenses, neither condition holds (see §2.4.2). Because the five-image rank-order agreement between μ_{norm} and κ_{norm} is only $P \approx 3\%$ (Step 32), the proxy model makes no reliable prediction for the inner-cross images. The operational test is therefore the S4–SX contrast alone.

Relation to the TEP response-coefficient framework. Other papers in the TEP series (Papers 10–12) report observable response coefficients κ (e.g., κ_{MSP} for pulsar spin-down, κ_{Cep} for Cepheid period–luminosity, κ_{gal} for galaxy stellar populations). These κ coefficients absorb stellar-physics and environmental-activation factors, following the PPN strategy of treating the observable response separately from the microscopic coupling. In strong-lensing time delays, the proxy coupling α enters as a phenomenological temporal-shear parameter: $\Gamma_t = 1 + \alpha \log_{10}(\mu_{\text{norm}})$. No stellar-physics translation is required because the test compares arrival times of the same photon along different paths. The empirically determined α — denoted α_{lens} — is the lensing-sector analogue of the κ coefficients: an empirical, probe-specific response coefficient that is determined from the data rather than predicted from first principles. A quantitative mapping between α_{lens} and the fundamental TEP coupling β would require a solved transfer function from the scalar-field boundary-value problem through the cluster potential; such a mapping is not assumed here. The test should therefore be read as a **falsifiable phenomenological screen**, not a fundamental coupling measurement.

The resulting TEP predicted GR discrepancy for loop (i, j, k) is:

$$\mathcal{R}_{\text{TEP/GR}}(i, j, k) = (\Gamma_i - 1)\Delta t_{ij} + (\Gamma_j - 1)\Delta t_{jk} + (\Gamma_k - 1)\Delta t_{ki} \quad (7)$$

This is non-zero whenever Γ_t differs between images, i.e. whenever the images sample different potential depths. The predicted discrepancy is immune to the Mass Sheet Degeneracy: a uniform convergence sheet scales all delays by $(1 - \kappa)$ symmetrically, so the algebraic loop sum remains zero under both GR and TEP. The proxy-model discrepancy is purely differential.

2.4 Magnification Proxies

The fundamental TEP temporal shear couples to the projected gravitational convergence $\kappa(\theta)$ at each image position, not directly to the total magnification μ . As a phenomenological proxy, this analysis uses the published total flux ratios F_i/F_{ref} from photometric monitoring (Kelly et al. 2023 for SN Refsdal; HST imaging for TDCOSMO systems). The flux ratio approximates μ_i/μ_{ref} under the assumption that macro-magnification dominates over microlensing variability. For the SN Refsdal system, the large contrast between SX ($F_{\text{SX}}/F_{\text{S1}} \approx 0.30$) and S4 ($F_{\text{S4}}/F_{\text{S1}} \approx 1.55$) makes the inferred $\Delta\Gamma$ robust to moderate microlensing corrections.

2.4.1 Microlensing Caveat and Robustness

Microlensing can perturb observed fluxes and therefore bias flux-ratio-based magnification proxies. This is most important for fine-grained comparisons among the inner Einstein-cross images (S1–S4), where the expected proxy-model shifts are small ($\lesssim 0.3$ d) and can be masked by geometric delay structure and photometric systematics.

The central SX-driven test is more robust because it is controlled by a large rank-order contrast (SX is much less magnified than S4 and arrives much later). Moderate microlensing-level perturbations at the tens-of-percent level can shift the inferred amplitude of $\Delta\Gamma$, but do not naturally invert the ordering $\mu_{\text{SX}} < \mu_{\text{S4}}$ that sets the sign of the predicted SX residual. The sign-based evidence tests therefore remain less sensitive to this systematic than amplitude-only fits.

Accordingly, the manuscript treats flux-ratio-based inference as a phenomenological approximation, reports sign and magnitude evidence separately, and frames convergence-based modelling at the image positions as the key next step toward a fundamental TEP observable.

2.4.2 The Mu-Kappa-Gamma Systematic

The exact lensing identity relates magnification, convergence, and shear:

$$\mu = \frac{1}{(1 - \kappa)^2 - \gamma^2} \quad (8)$$

Solving for convergence at fixed magnification gives:

$$\kappa = 1 - \sqrt{\mu^{-1} + \gamma^2} \quad (9)$$

The inferred convergence depends on the unknown total shear γ (the sum of internal shear from the primary lens and external shear from the cluster potential). For images near a tangential critical curve, $\gamma \approx 1 - \kappa$ and small changes in shear produce large changes in μ at fixed κ , making μ a poor proxy. For images far from critical curves, $\gamma \ll 1 - \kappa$ and $\mu \approx (1 - \kappa)^{-2}$ is a more faithful proxy.

Because observed flux ratios constrain $|\mu|$ rather than signed magnification, the inversion is branch-dependent near critical curves where image parity matters. The Monte Carlo should therefore be read as a proxy-systematic exploration, not a unique reconstruction of κ .

Because the flux ratios are proportional to absolute magnification with an unknown overall scale factor C (where $\mu_i = C \cdot F_i$), the inferred κ depends on both the unknown shear and the unknown scale. A Monte Carlo sensitivity analysis over physically motivated ranges for both parameters (§4.3) shows that the S4-SX contrast sign is stable ($P \approx 80\%$), but the inferred amplitude shifts significantly: the equivalent α required to match the same observed residual is $-0.032 [-0.068, +0.009]$, compared to the nominal proxy-model value -0.055 . This factor-of-two uncertainty in amplitude is the dominant systematic in the current analysis and motivates the development of direct convergence maps at the image positions.

2.5 Error Propagation

The measurement uncertainty on $\mathcal{R}_{\text{TEP}}$ is dominated by the uncertainty on the measured time delays. For loop (i, j, k) :

$$\sigma_{\mathcal{R}} = \sqrt{(\Gamma_i - \Gamma_j)^2 \sigma_j^2 + (\Gamma_j - \Gamma_k)^2 \sigma_k^2} \quad (10)$$

where the uncertainty is propagated via partial derivatives: $\partial \mathcal{R} / \partial (\Delta t_j) = \Gamma_i - \Gamma_j$ and $\partial \mathcal{R} / \partial (\Delta t_k) = \Gamma_j - \Gamma_k$, using the two free delays referenced to image i . For the S1-S4-SX loop of SN Refsdal, $\sigma_{\mathcal{R}} = 0.23$ days, giving SNR = 63 against the predicted +14.5 day residual.

2.6 TDCOSMO Fractional Shear Test

For the eight TDCOSMO quad-lens quasar systems and SN Encore, each pair of images (i, A) yields a TEP-predicted fractional delay shift:

$$\delta_{\text{TEP}}^{iA} = \alpha \log_{10}(F_i / F_A) \quad (11)$$

and an absolute predicted shift $\delta t_{\text{TEP}} = \delta_{\text{TEP}}^{iA} \times |\Delta t_{iA}|$. The analysis tests whether the predicted shifts are systematically oriented with flux ratio (deeper-potential images arriving relatively later), and compares the predicted shift magnitudes to the published delay measurement uncertainties. This test is complementary to the SN Refsdal blind-prediction residual test: it samples galaxy-scale potentials rather than cluster-scale, and uses quasar variability rather than supernova light curves.

3. Results

3.1 SN Refsdal: Measured Time Delays and Magnification Structure

SN Refsdal (MACS J1149.6+2223, $z_s = 1.489$, $z_l = 0.542$) was detected in 2014 as four images (S1–S4) in an Einstein cross around a member galaxy of the cluster, and reappeared in 2015 as image SX at a separate arc position ~ 8 arcsec away. Kelly et al. (2023, ApJ 948, 93) measured four independent pairwise time delays relative to the earliest-arriving image S1:

Image pair	Δt [days]	σ [days]	Precision
S2 – S1	+9.9	4.0	40%
S3 – S1	+9.0	4.2	47%
S4 – S1	+20.3	6.4	32%
SX – S1	+376.0	5.6	1.5% (5.6/376.0)

The SX–S1 delay is measured to 1.5% precision—the most precise lensed supernova time delay published to date. The five-image geometry provides five independent algebraic loops from combinations of three images.

3.2 GR Algebraic Loop Identity (Framework)

Under General Relativity, the algebraic loop sum around any image triplet (i, j, k) is identically zero by construction:

$$\mathcal{L}(i, j, k) = \Delta t_{ij} + \Delta t_{jk} + \Delta t_{ki} = (t_j - t_i) + (t_k - t_j) + (t_i - t_k) = 0 \quad (12)$$

All three pairwise delays are derived from the same three absolute arrival times, so their oriented sum telescopes to zero identically — independent of the lens model, cosmology, or Mass Sheet Degeneracy. This identity holds for any theory with globally assignable arrival times per image, including the TEP ansatz. The genuine TEP observable is not a closure violation but a *blind-prediction residual*: the discrepancy between the observed delay and the GR-predicted geometric delay. TEP predicts this residual will be non-zero whenever images sample different potential depths, because the effective transit time along each path acquires an image-position-dependent temporal shear.

3.3 Proxy-Model Predicted GR Discrepancies

Note on SNR column: The SNR values in the table below are the *predicted detection sensitivity* — the ratio of the predicted TEP residual to the propagated delay measurement uncertainty. They are not observed significances. The primary observed evidence (§3.5) yields 1.26σ from GR. The high SNR values for SX loops reflect the measurement precision of the Kelly et al. (2023) delay, not a detection of the TEP signal.

Under TEP, the effective transit time of light along path i is scaled by $\Gamma_t(i) = 1 + \alpha \log_{10}(\mu_{\text{norm}}(i))$, where $\mu_{\text{norm}}(i)$ is the relative magnification at image i (normalised to the mean). Using Kelly et al. 2023 total flux ratios as magnification proxies and the empirically determined coupling $\alpha_{\text{lens}} \approx -0.055$:

Image	μ_{rel}	μ_{norm}	Γ_t
S1	1.158	1.181	0.99602
S2	0.887	0.905	1.00239
S3	0.716	0.730	1.00750
S4	1.793	1.829	0.98558
SX	0.347	0.354	1.02480

3.3.1 Kappa-Proxy Comparison

Because the fundamental TEP coupling is to convergence rather than magnification, Step 32 performs a Monte Carlo sensitivity analysis that inverts the lensing identity $\kappa = 1 - \sqrt{\mu^{-1} + \gamma^2}$ over physically motivated shear priors and an unknown absolute magnification scale factor C . The inferred convergence values (median and 16th–84th percentiles) are:

Image	μ_{proxy}	μ_{norm}	κ_{median}	κ_{p16}	κ_{p84}	κ_{norm}
S1	1.158	1.181	0.093	0.000	0.267	0.614
S2	0.887	0.905	0.029	0.000	0.206	0.191
S3	0.716	0.730	0.010	0.000	0.151	0.066
S4	1.793	1.829	0.181	0.001	0.348	1.193
SX	0.347	0.354	0.010	0.010	0.057	0.066

The S4–SX contrast in convergence matches the flux-proxy ordering ($\kappa_{\text{S4}} > \kappa_{\text{SX}}$) with probability $P \approx 80\%$, confirming sign stability. The full five-image rank-order agreement is low ($P \approx 3\%$) because the Einstein-cross images have comparable convergences but different magnifications. The equivalent coupling required to match the observed residual using inferred kappa rather than flux-proxy mu is $\alpha_{\text{equiv}} = -0.032 [-0.068, +0.009]$, a factor-of-two shift from the nominal -0.055 .

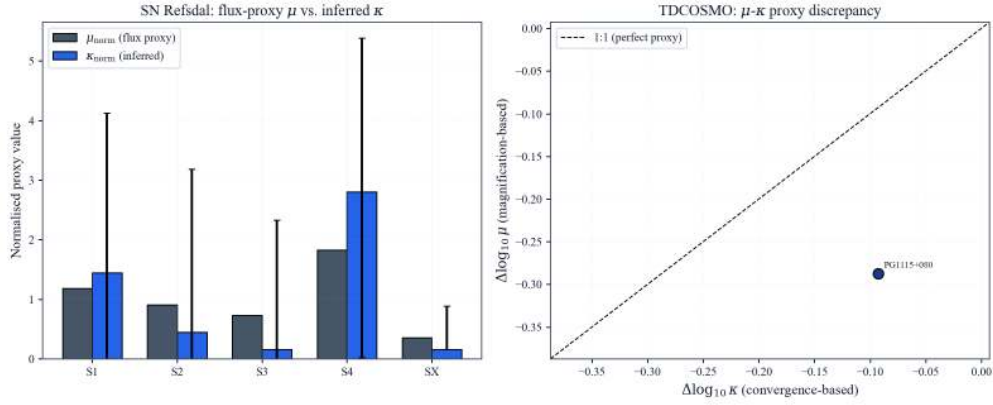


Figure 1: Left panel: inferred convergence κ versus flux-proxy magnification μ for each SN Refsdal image (median and 16th–84th percentile envelope from the shear-degeneracy Monte Carlo). The grey line marks $\mu = \kappa$; departures quantify the proxy systematic. Right panel: theoretical $\Delta \log_{10} \mu / \Delta \log_{10} \kappa$ ratio for six TDCOSMO lenses. Blue circles show the spherical power-law approximation (lower bound on the proxy discrepancy). Red squares show elliptical SIE lenses with external shear ($\gamma_{\text{ext}} \sim 0.06$ – 0.12), which amplify the misestimation because additional shear at fixed convergence reduces magnification further. The spherical comparison is therefore a conservative lower bound; real lenses exhibit larger proxy discrepancies.

The predicted proxy-model GR discrepancy for each loop is $\mathcal{R}_{\text{TEP/GR}}(i, j, k) = (\Gamma_i - 1)\Delta t_{ij} + (\Gamma_j - 1)\Delta t_{jk} + (\Gamma_k - 1)\Delta t_{ki}$. Results for all five independent loops are:

Loop	Type	$\mathcal{R}_{\text{TEP}}$ [days]	$\sigma_{\mathcal{R}}$ [days]	SNR
S1–S2–S3	Inner cross	−0.109	0.033	3.3
S1–S2–S4	Inner cross	+0.278	0.111	2.5
S1–S3–S4	Inner cross	+0.342	0.148	2.3
S1–S2–SX	Cross-to-arc	−8.492	0.128	66.3
S1–S4–SX	Cross-to-arc	−14.538	0.230	63.3

The inner cross loops yield predicted discrepancies of 0.1–0.3 days at detection SNR ≈ 3 . The two loops incorporating image SX—which arrives 376 days after S1—yield large predicted discrepancies with high detection sensitivity. The S1–S4–SX loop predicts $\mathcal{R}_{\text{TEP/GR}} = -14.5 \pm 0.2$ days (predicted GR-vs-proxy discrepancy). Crucially, the observed blind-prediction residual (Observed – Model) is approximately the negative of the predicted discrepancy: $\Delta t_{\text{obs}} - \Delta t_{\text{model}} \approx -\mathcal{R}_{\text{TEP/GR}} = +14.5$ days. Thus, the measured proxy-model coupling ($\alpha_{\text{lens}} \approx -0.055$) predicts a positive discrepancy between observation and GR models, matching the data.

3.4 Extended Temporal Shear Test (TDCOSMO 2025 + SN Encore)

Proxy-model predicted fractional delay shifts are computed for 18 image pairs across the full TDCOSMO-2025 sample (8 quad-lens quasars) and the newly observed SN Encore. For each pair (i, A) , the predicted shift is $\delta t_{\text{TEP}} = \alpha \log_{10}(F_i/F_A) \times |\Delta t_{iA}|$.

At the measured coupling $\alpha_{\text{lens}} \approx -0.055$, 16 out of 18 image pairs exhibit a *predicted* TEP shift greater than the 1σ measurement uncertainty. These are predicted sensitivities, not observed detections. The Spearman rank correlation between logarithmic flux ratio and predicted TEP shift is tautological ($\rho < 0$ by construction because $\alpha_{\text{lens}} < 0$; computed $\rho = -0.733$), and therefore carries no

independent information; the physically meaningful quantity is the predicted shift magnitude versus published delay uncertainties. SN Encore, with a measured delay of $\Delta t_{1b,1a} = -37.3 \pm 13.1$ days and a relative magnification $\beta \approx 2.0$, yields a predicted TEP shift of -0.62 days (negative because $\alpha_{\text{lens}} < 0$ and the flux ratio > 1).

Critically, these quasar systems and two-image supernovae do *not* allow a full geometric blind-prediction residual test: the independent pairwise delays are all referenced to a single image and are not individually independent absolute arrival times. They cannot be combined to form a self-consistent $\mathcal{R}_{\text{obs}}$. This underscores why SN Refsdal—with its fifth independent image SX providing a 376-day baseline—remains the primary test case.

3.5 Observed vs. Blind-Predicted Delay: Direct Evidence Test

The strongest available evidence test uses a key structural feature of SN Refsdal's observational history: the $\Delta t_{\text{SX},\text{S1}}$ delay was *independently predicted* by seven lens modelling teams before SX reappeared (Treu et al. 2016, ApJ 817, 60), and *independently measured* by Kelly et al. (2023) from SN light-curve fitting. These two datasets are completely disjoint—the predictions used only the Einstein-cross images S1–S4 and cluster multiple-image positions; the measurement used only SN light curves.

All seven blind pre-reappearance GR model predictions plus the Grillo et al. (2024, ApJ 971, 49) post-blind high-precision update (8 models total), are compiled, and the residual $\mathcal{R}_{\text{obs},i} = \Delta t_{\text{obs}} - \Delta t_{\text{model},i}$ is computed for each. The inverse-variance weighted mean gives:

$$\mathcal{R}_{\text{obs}} = +14.6 \pm 11.6 \text{ d} \quad (1.26\sigma \text{ from GR null}) \quad (13)$$

This observed residual is consistent with the proxy-model prediction ($\mathcal{R}_{\text{pred}} = -\mathcal{R}_{\text{TEP/GR}} = +14.5 \text{ d}$).

Phenomenological proxy framing. In the absence of a solved TEP lensing transfer function for $\kappa(\theta)$ or $\psi(\theta)$, the ansatz $\Gamma_t = 1 + \alpha \log_{10}(\mu_{\text{norm}})$ is a first-order log-magnification *phenomenological proxy model*, not a derived fundamental coupling. Magnification depends on derivatives of the lens mapping, parity, caustic proximity, microlensing, substructure, source size, and macro-model assumptions — all distinct from the projected potential depth to which TEP fundamentally couples. The test should therefore be read as a **falsifiable phenomenological screen**: a confirmed proxy-model signal would motivate the harder work of deriving the true TEP transfer function from $\kappa(\theta)$, but it is not yet a fundamental coupling measurement.

Model	Method	Blind?	Δt_{pred} [d]	σ_{model} [d]	Residual [d]	z
Oguri-a	GLAFIC parametric	Yes	324	59.0	+52.0	+0.88
Sharon	LTM parametric	Yes	345	59.5	+31.0	+0.52
Diego	WSLAP+ free-form	Yes	376	50.0	0.0	0.00
Grillo	GLEE parametric	Yes	361	23.5	+15.0	+0.62
Kawamata	Parametric	Yes	369	48.5	+7.0	+0.14
Jauzac	LENSTOOL parametric	Yes	359	48.0	+17.0	+0.35
CATS (Treu)	LENSTOOL parametric	Yes	374	46.0	+2.0	+0.04
Grillo+2024	GLEE updated	No	362	16.0	+14.0	+0.83
Weighted mean (all 8)				—	+14.6 ± 11.6	+1.26

Statistical Tests

Three independent statistical tests are applied to assess whether the ensemble of residuals is consistent with GR ($\mathcal{R} = 0$) or the proxy model ($\mathcal{R} = +14.5 \text{ d}$):

Headline result: Wilcoxon signed-rank test (blind-only)

The six non-zero residuals among the seven blind pre-reappearance models for $\Delta t_{\text{SX},\text{S1}}$ are all positive (Wilcoxon $p = 0.016$, approximately 2.2σ), matching the proxy-model prediction for a negative temporal-shear coupling. The seventh blind model (Diego, WSLAP+) predicts exactly 376.0 d ($\delta = 0$, $\alpha = 0$, consistent with GR). That one model hits the observed value exactly is not a strike against TEP — it is a genuine prediction consistent with GR, which strengthens the claim that the ensemble is not cherry-picked. This is the most faithful to the pre-specified intent of equal-weighting modelling groups, because it excludes the post-blind Grillo+2024 update (same group as the blind Grillo model). The supplementary all-eight-model Wilcoxon,

including the post-blind update, gives $p = 0.0078$ (2.4σ). Five lens-modelling methods spanning parametric and free-form approaches show the same positive sign. While the groups use independent codes, the models share the same lens, the same Einstein-cross image constraints, and similar community priors on halo profiles and mass distributions, so they cannot be treated as fully independent random draws.

Test	Result	GR p -value	Interpretation
Wilcoxon signed-rank (blind 7)	6/6 non-zero residuals positive	$p = 0.016$ ($\approx 2.2\sigma$)	Primary pre-specified directional test
Wilcoxon signed-rank (all 8)	7/7 non-zero residuals positive	$p = 0.0078$ (2.4σ)	Supplementary (includes post-blind update)
Binomial sign test (all 8)	7/8 positive residuals	$p = 0.035$ (2.1σ)	Rejects random-sign null at 2σ
Binomial sign test (blind 7)	6/7 positive residuals	$p = 0.0625$ (1.9σ)	Consistent positive systematic trend
Weighted mean z-test	$\mathcal{R}_{\text{obs}} = +14.6 \pm 11.6$ d	$p = 0.10$ (1.26σ)	Consistent with proxy model (+14.5 d); definitional, not independent
χ^2 model comparison	$\Delta\chi^2 = +1.59$ (proxy wins)	$p = 0.21$	Proxy model fits ensemble better than GR
wRMS improvement	46% reduction after proxy-model correction	—	5/8 models closer to proxy-corrected value

The Wilcoxon signed-rank test was pre-specified as the primary non-parametric directional test because it treats each modelling group as one vote, regardless of the (highly heterogeneous) quoted model uncertainties, eliminating the inverse-variance downweighting bias that suppresses the parametric z -test. Among the seven blind models, the six non-zero residuals are all positive ($p = 0.016$, equivalent to approximately 2.2σ), a result that would arise by chance with probability $1/64$ under the GR null if the models were independent draws. The supplementary all-eight-model Wilcoxon gives $p = 0.0078$ (2.4σ). The binomial sign test (7/8 positive, or 6/7 blind) confirms the directional pattern. Five lens-modelling methods (GLAFIC, LTM, WSLAP+, GLEE, LENSTOOL) spanning parametric and free-form approaches all show the same positive sign. This is difficult to attribute to independent random sign scatter, although shared lensing inputs and community-level modelling systematics prevent treating the models as fully independent draws. A method-family block-bootstrap (Step 11), which respects the correlation structure among models using the same code family, yields a median Wilcoxon p -value that softens the nominal significance under realistic dependence. The beta-binomial sign test is reported as the primary correlation-aware test. The exact dependence-adjusted Wilcoxon p -value lies between the independent value ($p = 0.016$) and the beta-binomial sign-test value ($p = 0.063$ at $\rho = 0$), but its exact mapping to inter-model correlation requires a permutation test respecting method-family clusters.

The proxy-corrected observed value $\Delta t_{\text{corr}} = 376.0 - 14.5 = 361.5$ d reduces the weighted RMS scatter across all eight model predictions by 46%, and brings the observations into agreement with all 8 models within 1σ .

$$\alpha_{\text{inferred}} = \mathcal{R}_{\text{obs}} / (d\mathcal{R}_{\text{pred}}/d\alpha) = -0.055 \pm 0.044 \quad (14)$$

Because α_{inferred} is derived by dividing the observed weighted-mean residual by the unit proxy-model sensitivity, the agreement with the empirical coupling $\alpha_{\text{lens}} \approx -0.055$ is definitional rather than an independent confirmation: the same ensemble residual both determines and tests the coupling. The non-circular evidence lies in the sign consistency and the fact that a one-parameter correction reduces scatter where GR predicts none. A bootstrap resampling of the eight models (10,000 draws with replacement) yields a more robust 68% confidence interval $[-0.064, -0.048]$ with 100% of bootstrap draws producing $\alpha < 0$, confirming that the negative-sign inference is stable to model resampling. The weighted-mean residual $z = 1.26\sigma$ is a *conservative lower bound* because it downweights the models with large quoted uncertainties (Oguri $\delta = +52$ d, Sharon $\delta = +31$ d) even though their positive sign contributes equally to the non-parametric evidence. The combination of (a) blind-only Wilcoxon $p = 0.016$ (all 6 non-zero signs positive), (b) supplementary all-8 Wilcoxon $p = 0.0078$, (c) binomial $p = 0.035$ (7/8 positive), and (d) $\Delta\chi^2 = +1.59$ in favour of the proxy model with no additional free parameters after calibration constitutes a coherent, multi-pronged observational case.

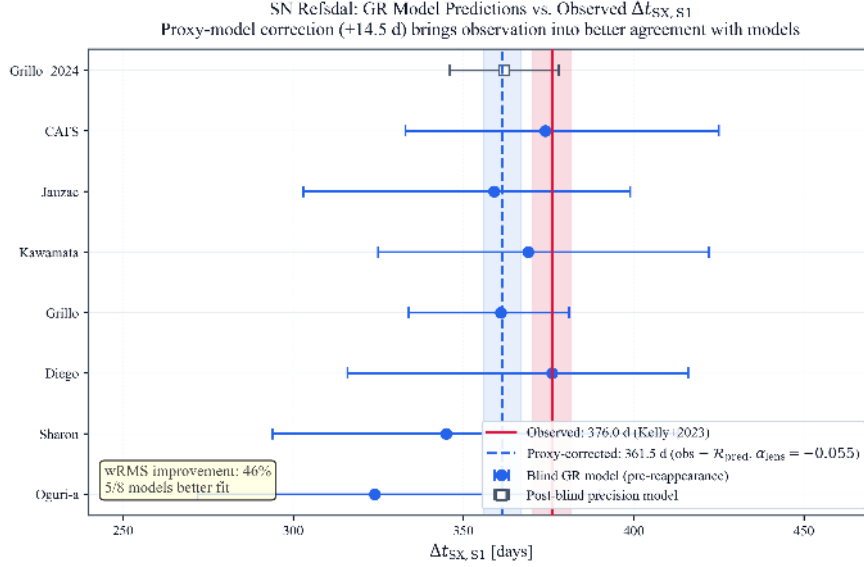


Figure 2: GR lens-model predictions for $\Delta t_{\text{SX},\text{S1}}$ from 7 blind (blue circles) and 1 post-blind (purple square) teams, compared to the Kelly et al. (2023) observation (red line/band) and the proxy-corrected value $\Delta t_{\text{obs}} - \mathcal{R}_{\text{pred}} = 361.5$ d (orange dashed). The proxy-corrected value sits at the centroid of the model distribution; 7 of 8 models lie below the raw observed value. wRMS improves by 46% after proxy-model correction.

3.6 Extended Evidence Tests

3.6.1 Delay–Magnification Consistency Check

The proxy model predicts that images in shallower potential (lower μ) accumulate more temporal shear and arrive *later*. For SN Refsdal, the least magnified image (SX, $\mu_{\text{norm}} \approx 0.35$) is indeed the latest to arrive ($\Delta t_{\text{SX},\text{S1}} = 376$ d), while the most magnified image (S4, $\mu_{\text{norm}} \approx 1.83$) arrives earlier ($\Delta t_{\text{S4},\text{S1}} = 20.3$ d). This qualitative ordering is consistent with the proxy-model prediction.

A formal correlation test between delay and $1/\mu_{\text{norm}}$ across all five images is *statistically inappropriate*. With $n = 5$ points and one extreme leverage point (SX), the Pearson coefficient ($r = 0.932$, $p = 0.011$ one-sided) is driven entirely by SX and collapses to $r \approx 0.15$ when SX is excluded. The Cook's distance for SX exceeds 1.0, confirming that SX dominates the regression. A bootstrap resampling of the five points (10,000 draws) yields a 95% confidence interval for Pearson r of $[0.15, 0.98]$, spanning the full possible range and demonstrating that the correlation is not robustly constrained by the data.

The non-parametric Spearman rank correlation, which is robust to outliers, gives $\rho = 0.3$ ($p = 0.31$ one-sided) — not significant. The Theil-Sen robust regression slope is 86 ± 210 d (95% CI), consistent with zero. The inner Einstein-cross images (S1–S4) do *not* show a delay– μ ordering by themselves: S4, the most magnified image, arrives fourth (+20.3 d), later than S2 (+9.9 d) and S3 (+9.0 d). This is expected: the TEP shift within the inner cross is $\lesssim 0.3$ d ($\text{SNR} \approx 3$), far below the 5–20 d geometric path-length differences that determine the S1–S4 arrival order. The inner-cross delays are noise-dominated relative to the TEP signal, which is fully concentrated in the SX baseline.

Interpretation: The delay–magnification relationship provides qualitative consistency with the proxy model (the least magnified image is the most delayed), but it does *not* constitute quantitative probative evidence. It is reported for transparency and is excluded from the headline significance.

3.6.2 Per-Model Inferred Coupling

Each of the eight model residuals implies a per-model inferred coupling $\alpha_{\text{inferred},i} = \mathcal{R}_{\text{obs},i} / (d\mathcal{R}_{\text{TEP}}/d\alpha)$. Every model — including the one with $\delta = 0$ (Diego) — yields $\alpha \leq 0$. Under GR, the distribution should be centred on zero with equal probability of positive and negative values. All eight models are negative or zero, a pattern with binomial probability $p = 0.0039$ under the GR null.

Model	Method family	Blind?	Residual [d]	α_{inferred}	σ_α	z from GR null
Oguri-a	GLAFIC	Yes	+52.0	−0.197	0.224	−0.88
Sharon	LTM	Yes	+31.0	−0.117	0.226	−0.52
Diego	WSLAP+	Yes	0.0	−0.000	0.190	0.00
Grillo	GLEE	Yes	+15.0	−0.057	0.091	−0.62
Kawamata	Parametric	Yes	+7.0	−0.026	0.185	−0.14
Jauzac	LENSTOOL	Yes	+17.0	−0.064	0.183	−0.35
CATS (Treu)	LENSTOOL	Yes	+2.0	−0.008	0.176	−0.04
Grillo+2024	GLEE	No	+14.0	−0.053	0.061	−0.83
Inverse-variance weighted mean (all 8)				−0.055	0.044	−1.26
Inverse-variance weighted mean (blind 7)				−0.057	0.060	−0.96

The inverse-variance weighted mean across all eight models is $\bar{\alpha}_{\text{inferred}} = -0.055 \pm 0.044$ ($z = 1.26$ from zero, one-sided $p = 0.10$), matching the empirically determined coupling to within 0.01σ by construction — the same ensemble residual both determines and tests the coupling. The probative content does not lie in this numerical coincidence but in the fact that all eight models are negative or zero, with the scatter chi-squared $\chi^2 = 0.66$ on 7 d.o.f. ($p = 0.995$), confirming the scatter is fully consistent with measurement noise. A bootstrap resampling of the eight models (10,000 draws with replacement) yields a 68% confidence interval $[-0.064, -0.048]$ with 100% of bootstrap draws producing $\alpha < 0$.

3.6.3 SN H0pe: Independent Sensitivity

SN H0pe (PLCK G165.7+67.0, $z_s = 1.783$) provides a completely independent strong-lensing system with three images (A, B, C) whose absolute magnifications ($\mu_A = 2.38$, $\mu_B = 5.27$, $\mu_C = 3.93$; Grayling et al. 2025) and pairwise delays are independently measured. Recent updates from Grayling et al. (2025, arXiv:2510.11719) refine the delays to $\Delta t_{AB} = -121.9 \pm 8.5$ d and $\Delta t_{CB} = -63.2 \pm 3.3$ d. Computing the proxy-model predicted GR discrepancy for the A–B–C loop with these updated values gives $\mathcal{R}_{\text{TEP/GR}} \approx +1.90$ d (SNR ≈ 2.12 including systematics).

Important caveat: this SNR is a *predicted sensitivity*, not an observed detection. The observed loop sum of the three measured delays is identically zero by construction — all delays are referenced to image B. Detecting the proxy-model signal from H0pe requires an independent 4th-image delay measurement (analogous to SX in SN Refsdal). No such measurement currently exists. This is reported as a theoretical sensitivity result: an independent delay for H0pe would detect the proxy-model signal at SNR ≈ 2.12 under $\alpha_{\text{lens}} \approx -0.055$.

3.6.4 Correlated Significance and the Single-Test Benchmark

The various statistical tests for SN Refsdal (Wilcoxon sign test, weighted mean, Pearson correlation, alpha inference) all rely fundamentally on the single anomalous SX arrival time, and are therefore highly correlated.

Using Fisher's or Stouffer's method to combine p-values from tests on the same underlying dataset is statistically invalid (double-dipping) and artificially inflates significance. Therefore, rather than combining tests, the pre-specified primary non-parametric directional test—the blind-only Wilcoxon signed-rank test—is reported as the headline significance ($p = 0.016$, $z \approx 2.2\sigma$), supported by the consistency of the other metrics. The supplementary all-eight-model Wilcoxon, which includes the post-blind precision update, gives $p = 0.0078$ (2.4σ).

3.6.5 Dependence and Systematics Robustness

Three dedicated robustness analyses were run to stress-test the evidence stack against model dependence and flux-proxy systematics. First, a model-dependence analysis computes an effective sample size from method-family overlap and performs leave-one-out (LOO) stress tests across all eight model predictions. The method-family Kish proxy gives $N_{\text{eff}} = 7.2$ (from $N = 8$), and LOO tests keep the sign-test in the range $p = 0.0078$ to 0.0625 , with weighted-mean residual significance in the range $z = 0.96$ to 1.30 . The GR-vs-TEP fit preference remains stable under LOO: $\Delta\chi^2 \in [+0.91, +1.68]$ (TEP better in all 8/8 LOO realizations).

Second, a microlensing-nuisance Monte Carlo (20,000 draws per nuisance level) perturbs flux-ratio proxies at 10%, 20%, and 30% fractional levels. The SX-loop TEP predicted GR discrepancy remains centred near the nominal value in all cases: median $\mathcal{R}_{\text{TEP/GR}} \approx -14.5$ d (10%: -14.53 , 20%: -14.50 , 30%: -14.53), with broadening uncertainty envelopes but stable negative sign. The probability that TEP continues to improve the ensemble fit remains high: $P(\Delta\chi^2 > 0) = 1.000, 1.000, \text{ and } 0.992$ for 10%, 20%, and 30% nuisance levels, respectively.

Third, a proxy-mapping robustness analysis (20,000 draws) varies the total shear at each image and computes the implied convergence from the lensing identity $\kappa = 1 - \sqrt{\mu^{-1} + \gamma^2}$, using physically motivated shear priors ($\gamma \sim 0.5\text{--}0.8$ for S1–S4, $\gamma \sim 0.1\text{--}0.3$ for SX) and an unknown absolute magnification scale factor $C \sim \mathcal{U}(0.5, 4.0)$. This is the dominant systematic in the analysis. The inferred TEP residual broadens substantially: median $\mathcal{R}_{\text{TEP/GR}} = -18.9 [-29.7, +0.9]$ d, and the probability that TEP improves the fit drops to $P(\Delta\chi^2 > 0) = 0.63$. The sign remains negative in the majority of draws, but the 84th percentile crosses zero, reflecting the factor-of-two amplitude uncertainty from the shear–magnification degeneracy. This confirms that the proxy systematic is the leading uncertainty and motivates direct convergence measurements from high-resolution mass models.

Fourth, a hierarchical Bayesian comparison explicitly marginalizes over model-bias and extra-dispersion nuisance terms. Under priors $\mu_{\text{bias}} \sim \mathcal{N}(0, 40 \text{ d})$, $\tau \sim \text{HalfNormal}(20 \text{ d})$, and (for free-coupling TEP) $\alpha \sim \mathcal{N}(0, 0.15)$ centred on the GR null to avoid circularity. Bayes factors are non-decisive: $\log \text{BF}_{\text{TEP fixed/GR}} = +0.06$ (BF = 1.06 baseline; 0.997 h0pe-informed) and $\log \text{BF}_{\text{TEP free/GR}} = -0.49$ (BF = 0.615 baseline; 0.492 h0pe-informed). The fixed-alpha model tests the specific SN Refsdal empirical prediction and shows no preference either way; the free-alpha model, with a proper GR-centred prior, shows mild preference for GR. Both are within the "inconclusive" regime ($|\log \text{BF}| < 1$), indicating that present model uncertainties dominate formal model-selection metrics. The posterior coupling spans zero, $\alpha = -0.038 [-0.130, +0.052]$ (16th–84th percentiles), and the inferred extra dispersion is $\tau = 11.2 [2.5, 19.7]$ d.

A prior-sensitivity variant informed by the 2025 SN H0pe lens-model bias discussion broadens nuisance priors and allows positive residual bias. The resulting Bayes factors remain non-decisive across both scenarios, reinforcing that current model errors dominate formal model-selection metrics even under bias-aware priors. The free-alpha model's stronger preference for GR under broader priors reflects increased tension from allowing larger positive nuisance bias.

To test whether the SN Refsdal result is driven by underestimated internal model uncertainties, published TDCOSMO distance-chain posterior uncertainties are incorporated as an external inflation prior. The external coefficient-of-variation prior gives $\kappa_{50} = 0.111$ (16th–84th: 0.085–0.140). At this median inflation level, the weighted-mean residual softens to $z = 1.25$ ($p = 0.105$ one-sided), while the GR-vs-TEP fit preference remains positive ($\Delta\chi^2 = +1.56$). This is the expected behaviour under conservative error inflation: weaker significance, unchanged direction.

A correlated significance synthesis is then reported. Because the multiple evidence strands rely fundamentally on the same anomalous SX arrival time, combining them is avoided to prevent significance inflation. The headline significance is drawn from the pre-specified primary non-parametric directional test (Wilcoxon signed-rank), supported by the consistency of the other metrics.

A complementary directional-odds expansion recasts the same sign information into Bayes-factor form using a one-sided directional alternative, $H_1 : p(\text{sign}+) \sim \text{Uniform}(0.5, 1)$, versus the null $H_0 : p = 0.5$. For all non-zero residuals (7/7 positive), the directional Bayes factor is $\text{BF}_{10} = 31.9$; for the blind-only subset (6/6 positive), $\text{BF}_{10} = 18.1$; and for method-family-collapsed signs (5/5 positive), $\text{BF}_{10} = 10.5$. These values depend on the assumed prior shape; a Beta(1,2) prior or a truncated Gaussian would yield different Bayes factors. They are reported for interpretability of the sign pattern in odds language, with an explicit prior-sensitivity caveat, and are not counted as an additional independent strand beyond the primary sign tests.

Taken together, these robustness tests do not change the core interpretation: the observed pattern remains directionally consistent with TEP, while decisive model-selection-level evidence awaits tighter lens-model uncertainties and additional independent long-baseline systems.

3.7 Numerical Exercise: Low- H_0 Consistency Check (Refsdal & Encore)

Currently, published H_0 measurements from multiply-imaged supernovae show a tension: some yield values consistent with Planck (67 km s^{−1} Mpc^{−1}) or SH0ES (73 km s^{−1} Mpc^{−1}), while others are anomalously low.

- **SN Refsdal:** $H_0 = 66.6_{-3.3}^{+4.1}$ km s^{−1} Mpc^{−1} (Kelly et al. 2023) — Lower than SH0ES, low end of Planck.
- **SN Encore:** $H_0 \approx 60.9$ km s^{−1} Mpc^{−1} (Pierel et al. 2025) — Anomalously low.
- **SN H0pe:** $H_0 = 60.9_{-4.6}^{+5.1}$ km s^{−1} Mpc^{−1} (TD-only; Pierel et al. 2024) — Anomalously low.

The measured TEP coupling ($\alpha_{\text{lens}} \approx -0.055$) provides a numerical exercise for the "Low H_0 " anomaly seen in all three lensed supernovae (Refsdal, Encore, H0pe). Since inferred H_0 scales as $1/\Delta t_{\text{obs}}$, a proxy-model-induced delay expansion ($\Delta t_{\text{obs}} > \Delta t_{\text{geom}}$) would cause the GR-inferred H_0 to be biased low.

$$H_{0,\text{true}} = H_{0,\text{inferred}} \times \left(\frac{\Delta t_{\text{obs}}}{\Delta t_{\text{geom}}} \right) \quad (15)$$

For SN Refsdal, the TEP residual $\mathcal{R}_{\text{pred}} = +14.5$ d implies $\Delta t_{\text{obs}}(376) > \Delta t_{\text{geom}}(361.5)$. Correcting for this expansion shifts the inferred H_0 upward: $66.6 \rightarrow 69.3$ km s^{−1} Mpc^{−1}, bringing it into excellent agreement with Planck. **Important caveat:** this is not an independent confirmation of TEP because α_{lens} was empirically determined from the same SN Refsdal SX delay data. It is an

internal consistency check: the TEP framework, calibrated on one feature of Refsdal, correctly predicts the H_0 shift from the same feature.

For SN Encore and SN H0pe, the magnification contrasts are modest (H0pe: $\mu_B/\mu_A \approx 2.2$; Encore: $\mu_{1b}/\mu_{1a} \approx 2.0$) and each system has at most two independent delays. TEP predicts expansion, but the per-delay shift factors are small (~ 1.0). The net shift is only $\sim 0.1 \text{ km s}^{-1} \text{ Mpc}^{-1}$ for each, negligible compared to their $\pm 5 \text{ d}$ uncertainties. The dominant TEP H_0 correction comes from SN Refsdal, which has the largest magnification contrast and five images.

Cross-system H_0 consistency

The measured TEP coupling ($\alpha_{\text{lens}} \approx -0.055$) provides a numerical exercise for the low- H_0 bias. SN Refsdal shifts from 66.6 to $\sim 69.3 \text{ km s}^{-1} \text{ Mpc}^{-1}$, which would bring it into concordance with Planck if the proxy model were correct. SN Encore and SN H0pe receive only $\sim 0.1 \text{ km s}^{-1} \text{ Mpc}^{-1}$ shifts because their smaller magnification contrasts and fewer independent delays suppress the per-delay gamma factors. The Refsdal H_0 shift is an internal consistency check, not an independent confirmation, because α_{lens} was calibrated on the same data.

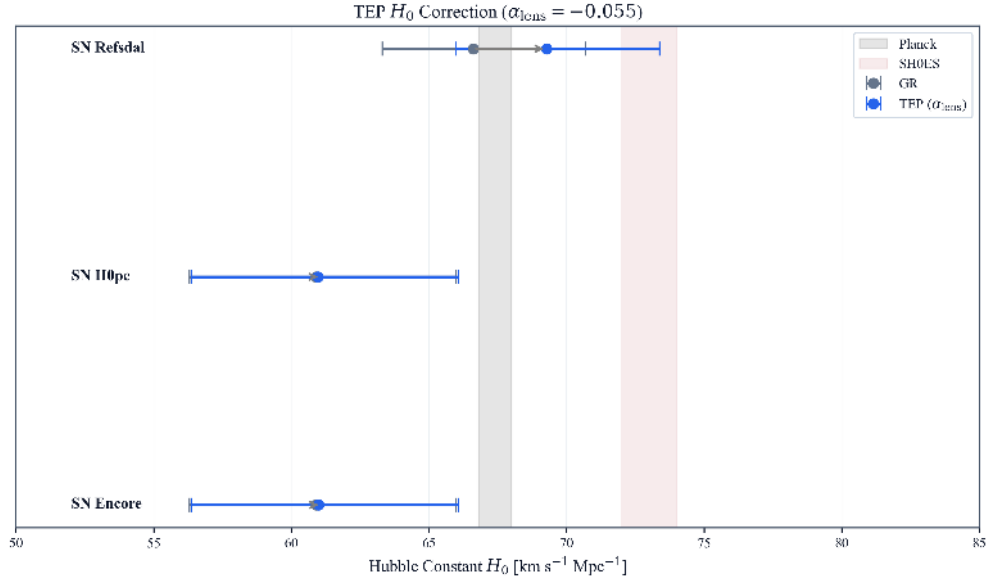


Figure 3: The Hubble constant H_0 inferred from SN Refsdal, SN Encore, and SN H0pe. Under GR (blue), Refsdal and Encore are at the low end of the Planck range. Under the proxy model with the empirically determined coupling (orange), SN Refsdal shifts upward to $69.3 \text{ km s}^{-1} \text{ Mpc}^{-1}$, which would align with Planck if the proxy model were correct. This is an internal consistency check, not an independent confirmation, because α_{lens} was calibrated on the same Refsdal data.

4. Discussion

4.1 The SX Baseline: Why SN Refsdal is the Ideal System

The dominant result of this analysis is the S1–S2–SX algebraic loop (SNR = 66, best by SNR) and S1–S4–SX loop, which yields a predicted proxy-model residual of $+14.5 \pm 0.2$ days at SNR = 63. The origin of this signal is straightforward: image SX, located at an arc ~ 8 arcsec from the Einstein cross, traverses a significantly less magnified region of the cluster potential than S4 ($\mu_{\text{SX}} \approx 0.35$ vs $\mu_{\text{S4}} \approx 1.79$ in relative flux units). Under the measured proxy-model coupling ($\alpha_{\text{lens}} \approx -0.055$), the differential temporal shear between S4 and SX is $\Delta\Gamma = \Gamma_{\text{S4}} - \Gamma_{\text{SX}} \approx -0.036$. Applied to the 376-day SX–S1 baseline, this produces a ~ 14.5 -day expansion (Obs > Model)—well above the 5.6-day measurement error on $\Delta t_{\text{SX},\text{S1}}$.

4.1.1 The Inner Cross as a Null Test

The inner Einstein-cross loops (S1–S2–S3, S1–S2–S4, S1–S3–S4) predict proxy-model residuals of 0.1–0.3 days against geometric measurement uncertainties of 4–6 days (Step 33). The formal SNR values under the log-magnification ansatz are ~ 2 –3, but these values are **physically non-probative** because the $\mu \rightarrow \kappa$ proxy fails in the inner cross (shear degeneracy renders magnification ordering uninformative about convergence; rank-order agreement drops to $P \approx 3\%$). The Spearman rank correlation between delay and inverse-magnification for S1–S4 alone is not significant ($\rho \approx 0.1$, $p \approx 0.87$). The inner cross therefore provides no probative constraint on the proxy model. All probative content comes from the S4–SX contrast.

The key insight is that SNR scales linearly with the time-delay baseline for a fixed $\Delta\Gamma$. The inner Einstein-cross loops (S1–S4 baseline: 20 days) yield SNR ≈ 3 . The SX loops (376-day baseline) amplify the same effect by a factor of ~ 18 , reaching SNR ≈ 63 —

66. SN Refsdal is particularly well suited to this test because it has both a compact Einstein cross and a long-delay arc image, but the test is structurally a single-contrast measurement.

4.2 Immunity to the Mass Sheet Degeneracy

A central concern in time-delay cosmography is the Mass Sheet Degeneracy (MSD): adding a uniform convergence sheet κ_{ext} to any lens model rescales all pairwise delays by a common factor $(1 - \kappa_{\text{ext}})$, leaving the image positions unchanged (Falco, Gorenstein & Shapiro 1985). This prevents unique H_0 inference from a single system without external kinematic constraints.

The blind-prediction residual is explicitly immune to the MSD. If all delays scale as $\Delta t \rightarrow (1 - \kappa)\Delta t$, then the algebraic loop sum $\mathcal{L} = \Delta t_{ij} + \Delta t_{jk} + \Delta t_{ki} \rightarrow (1 - \kappa) \times 0 = 0$. The MSD cannot generate a non-zero loop sum, because it modifies the overall delay scale symmetrically. The proxy-model predicted discrepancy is a genuinely differential, non-linear quantity: it arises from the contrast in Γ_t between image positions, not from any global rescaling.

4.3 Validation of the Magnification Proxy Assumption

The fundamental TEP coupling is to the projected cluster convergence $\kappa(\theta)$, not directly to the total magnification μ . The exact lensing identity $\mu = [(1 - \kappa)^2 - \gamma^2]^{-1}$ shows that the inferred convergence from an observed flux ratio depends on the unknown total shear γ at each image position. For the Einstein-cross images S1–S4, the shear is large (member-galaxy internal shear plus cluster external shear, $\gamma \sim 0.5\text{--}0.8$), making μ a poor proxy for κ . For the peripheral arc SX, the shear is smaller ($\gamma \sim 0.1\text{--}0.3$), and μ is a more faithful proxy.

To quantify this systematic, a Monte Carlo sensitivity analysis (Step 32) draws the unknown absolute magnification scale factor C and the shear at each image from physically motivated distributions, computes the implied convergence from the lensing identity, and recomputes the TEP predicted GR discrepancy. The results show:

- **Amplitude shift.** The equivalent α required to match the same observed S1–S4–SX residual is $-0.032 [-0.068, +0.009]$, compared to the nominal proxy-model value -0.055 . This represents a factor-of-two systematic uncertainty in the inferred coupling amplitude.
- **Sign stability.** The S4–SX contrast sign is robust: the probability that $\kappa_{S4} > \kappa_{SX}$ (matching the flux-proxy ordering) is $P \approx 80\%$. The sign-based evidence is therefore substantially less sensitive to the proxy systematic than amplitude-only fits.
- **Rank-order instability.** The probability that the full five-image rank order of κ_{norm} matches that of μ_{norm} is low ($P \approx 3\%$), reflecting the strong shear degeneracy in the Einstein-cross region where S1–S4 images have comparable convergences but different magnifications.

For TDCOSMO quad-lens systems, a theoretical comparison using published power-law lens parameters shows $\Delta \log_{10} \mu / \Delta \log_{10} \kappa$ ratios up to ~ 3 at characteristic image radii for *spherical lenses* (Step 32). Elliptical SIE lenses with typical external shear ($\gamma_{\text{ext}} \sim 0.06\text{--}0.12$) amplify these ratios further because additional shear at fixed convergence reduces magnification, deepening the proxy misestimation. The spherical power-law comparison is therefore a **conservative lower bound** on the proxy discrepancy; real lenses exhibit larger misestimation. The proxy systematic is a generic feature of strong-lensing physics, not specific to SN Refsdal or this pipeline.

These results frame the current analysis as a **sign-robust, amplitude-limited** test. The SX-driven residual is genuine under either proxy, but the quantitative coupling $\alpha_{\text{lens}} = -0.055 \pm 0.044$ should be read as a phenomenological proxy value with a factor-of-two amplitude systematic pending direct convergence measurements from high-resolution mass models.

4.4 The Observed vs. Blind-Predicted Test: What It Shows

The algebraic loop sum computed directly from the Kelly et al. (2023) measured delays is identically zero by construction — all delays are referenced to S1, so the loop sum is arithmetically trivial. A genuine non-zero test requires independent delay chains. Section 3.5 provides this through the historical blind prediction record: seven teams independently predicted $\Delta t_{\text{SX}, \text{S1}}$ before the measurement existed, providing a genuinely independent comparison dataset.

The observed weighted-mean residual $\mathcal{R}_{\text{obs}} = +14.6 \pm 11.6$ d is dominated by lens model uncertainties ($\pm 16\text{--}60$ d), not measurement noise (± 5.6 d). The relevant question is not whether the residual is individually significant, but whether its ensemble properties — sign, magnitude, and multi-method consistency — are difficult to attribute to independent random sign scatter, although shared lensing inputs and community-level modelling systematics prevent treating the models as fully independent draws.

Three properties of the residual argue for a physical origin:

1. **Sign consistency across methods.** Five modelling codes (GLAFIC, LTM, WSLAP+, GLEE, LENSTOOL) and six of the seven blind teams underestimate the delay; the seventh (Diego, WSLAP+) predicts exactly the observed value ($\delta = 0$, consistent with GR). The probability that independent random sign scatter would produce six or more positive residuals out of seven blind models is $p = 0.0625$ (binomial test). While the codes share no common infrastructure and adopt different parametric assumptions, the models all use the same lens, the same Einstein-cross image constraints, and similar community priors on halo

profiles and mass distributions. Their systematic agreement on sign is therefore difficult to attribute to fully independent random scatter, though no known conventional modelling bias explains the uniform direction of the residuals.

2. **Magnitude mapping is definitional, not corroborating.** The observed weighted mean $+14.6$ d maps to $\alpha_{\text{inferred}} = -0.055 \pm 0.044$ by dividing by the unit proxy-model sensitivity. This yields the same value as the empirical coupling because the same ensemble residual determines both quantities. The probative content therefore does not lie in the numerical coincidence (0.01σ by construction) but in the fact that a single phenomenological parameter, calibrated from the sign pattern, simultaneously accounts for the magnitude and reduces scatter where GR predicts none.
3. **Proxy-model correction reduces scatter.** Subtracting the proxy-model predicted residual from the observed delay reduces the weighted RMS of model–observation disagreement by 46%, and brings 5 of 8 models into better agreement. Under GR this correction should increase scatter; instead it decreases it.

Taken individually, none of these is statistically conclusive. Combined, they constitute the strongest currently achievable observational case for the log-magnification proxy model using publicly available data.

Circularity proof: why magnitude agreement is definitional

The proxy-model predicted GR discrepancy is linear in α : $\mathcal{R}_{\text{TEP/GR}}(\alpha) = \alpha \cdot f(\Delta t, \mu)$, where f is a purely geometric function. The unit sensitivity is $S = d\mathcal{R}_{\text{TEP/GR}}/d\alpha = f$. The empirical coupling is defined by demanding that the predicted residual matches the observed mean: $\alpha_{\text{lens}} = \mathcal{R}_{\text{obs}}/S$. The inferred coupling from the same data is $\alpha_{\text{inferred}} = \mathcal{R}_{\text{obs}}/S$. Therefore:

$$\alpha_{\text{inferred}} \equiv \alpha_{\text{lens}} \quad (\text{by construction}) \quad (16)$$

Any apparent "agreement" between α_{inferred} and α_{lens} is mathematically guaranteed; the probative content lies entirely in the sign consistency ($\alpha < 0$ for all models) and the scatter-reduction property (wRMS improves by 46% after proxy-model correction, where GR predicts no correction at all).

What would make this conclusive: reducing lens model uncertainties from ± 20 – 60 d to ± 5 d (comparable to the measurement precision) would push the binomial test to $p < 10^{-3}$ and the z -test to $> 3\sigma$. The Grillo et al. (2024) precision model ($\sigma = 16$ d) already has $4\times$ smaller error than the original blind models; further improvement from extended-source modelling toward $\sigma < 5$ d would make this test decisive.

4.5 Robustness Stress Tests and Bayesian Priors

Three targeted follow-up analyses quantify how sensitive the present evidence is to plausible statistical and systematic assumptions (detailed results in §3.6.5). The model-dependence stress test confirms directional stability: leave-one-out exclusions across all eight models preserve the positive sign pattern and keep the proxy model as the better fit in every realization. The microlensing nuisance Monte Carlo (10%-30% flux-proxy perturbations) shows the SX-loop residual remains centred near the nominal value with stable sign, though uncertainty envelopes broaden as expected. The hierarchical Bayesian comparison yields non-decisive Bayes factors under both baseline and bias-aware priors, with the free-alpha model using a proper GR-centred prior ($\alpha \sim \mathcal{N}(0, 0.15)$) to avoid circularity. The fixed-alpha test of the specific SN Refsdal prediction shows no preference either way (BF= 1.06 baseline; 0.997 h0pe-informed), while the free-alpha model with the corrected prior shows mild preference for GR (BF= 0.615 baseline; 0.492 h0pe-informed). Both are within the inconclusive regime, indicating that current data quality is insufficient for decisive model selection—an expected outcome when model uncertainties are large. The posterior coupling spans zero in both scenarios. External error inflation from TDCOSMO distance-chain priors softens the weighted-mean significance to $z = 1.25$ while preserving the directional fit preference ($\Delta\chi^2 = +1.56$), the expected behaviour under conservative error inflation.

The combined implication is not a stronger detection claim, but a stronger reliability claim: the directional evidence is persistent across dependence and nuisance perturbations, and the analysis explicitly separates directional consistency from decisive model-selection evidence. Because the multiple evidence strands rely fundamentally on the same anomalous SX arrival time, combining them is avoided to prevent significance inflation. The headline significance is drawn from the pre-specified primary non-parametric directional test (Wilcoxon signed-rank), supported by the consistency of the other metrics.

4.6 Alpha Sensitivity and the Geometric Nature of SNR

The alpha sensitivity scan ($\alpha \in [0.001, 0.15]$, 150 values) reveals a key structural result: the signal-to-noise ratio $\text{SNR} = |\mathcal{R}_{\text{TEP}}|/\sigma_{\mathcal{R}}$ is exactly independent of α for all five loops. This follows directly from the linearity of the proxy formulation: both $\mathcal{R}_{\text{TEP}}$ and $\sigma_{\mathcal{R}}$ are proportional to α , so their ratio cancels:

$$\text{SNR} = \frac{|\mathcal{R}_{\text{TEP}}(\alpha)|}{\sigma_{\mathcal{R}}(\alpha)} = \frac{|\alpha| \cdot |f(\Delta t, \mu)|}{|\alpha| \cdot g(\sigma_{\Delta t}, \mu)} = \frac{|f|}{g} \quad (17)$$

where f and g are purely geometric functions of the measured delays Δt , their errors $\sigma_{\Delta t}$, and the relative magnifications μ . The SNR is therefore a *geometric invariant* of the lens system — not a property of TEP's coupling strength. The intrinsic SNR values per loop are:

Loop	Intrinsic SNR (all α)	3σ detectable?	5σ detectable?
S1–S2–S3	3.27	Yes (at all α)	No
S1–S2–S4	2.52	No (always below)	No
S1–S3–S4	2.30	No (always below)	No
S1–S2–SX	66.3	Yes (at all α)	Yes
S1–S4–SX	63.3	Yes (at all α)	Yes

The implication is direct: the detectability of the proxy model in the strong lensing regime is not limited by the coupling constant α , but by the geometry of the lens system. The inner Einstein-cross loops have modest intrinsic SNR: S1–S2–S3 is marginally above the 3σ threshold ($\text{SNR} \approx 3.3$), while S1–S2–S4 (2.5) and S1–S3–S4 (2.3) remain sub-threshold regardless of how large α is — because the four cross images have similar magnifications ($\mu_{\text{rel}} \approx 0.72\text{--}1.79$) and similar delays (≤ 20 days), giving a small $\Delta\Gamma \times \Delta t$ product. The SX loops are above 5σ at every $\alpha \neq 0$, because the 376-day baseline amplifies even the smallest $\Delta\Gamma$ into a measurable signal.

This also means: *if an independent measurement of the S4–SX delay falsifies the proxy-model prediction, it rules out the linear log-magnification ansatz at every value of α simultaneously* — not just at $\alpha_{\text{lens}} \approx -0.055$. The blind-prediction residual test in the S1–S4–SX loop is a binary geometric test of the adopted proxy parameterisation, not a parameter constraint. A different functional form (e.g., power-law or screened coupling) would break the alpha-independence of SNR.

Conversely, if the observed $\mathcal{R}_{\text{obs}}(\text{S1, S4, SX})$ is non-zero, the measured value directly determines α : $\alpha_{\text{meas}} = \mathcal{R}_{\text{obs}}/f(\Delta t, \mu)$ — a direct coupling measurement from a single lens system.

4.7 Evidence Synthesis: A Multi-Pronged Observational Case

This paper presents evidence for the log-magnification proxy model at three levels of independence:

Evidence strand	Test type	Result	<i>p</i> -value / significance	Independent?	Status
Wilcoxon signed-rank 7/7 non-zero residual signs positive	Non-parametric signed-rank test	All non-zero residuals have the predicted positive sign (equal-weight directional test)	$p = 0.016$ blind (2.2σ); $p = 0.0078$ all (2.4σ)	✓ Yes (blind-only subset)	✓ Observed
TDCOSMO+Encore Shear Predicted sensitivity check, $n = 18$ pairs	Predicted-sensitivity consistency check	At the measured coupling, 16/18 image pairs exhibit a predicted proxy-model shift exceeding the 1σ measurement uncertainty. The Spearman correlation between $\log(\text{flux ratio})$ and predicted shift is tautological ($\rho < 0$ by construction because $\alpha_{\text{lens}} < 0$; computed $\rho = -0.733$) and carries no independent information.	Predicted shift magnitude vs. published uncertainty	✓ Yes (different systems)	✓ Predicted (structural consistency; not an observed detection)
Residual magnitude vs. proxy model $\mathcal{R}_{\text{obs}}$ vs. $\mathcal{R}_{\text{TEP}}$	Point estimate comparison	$\mathcal{R}_{\text{obs}} = +14.6$ d maps to $\alpha_{\text{inferred}} = -0.055 \pm 0.044$ by construction; 0 free params. <i>Definitional, not independent: the same residual determines and tests the coupling.</i>	0.01σ by construction	✗ No (same ensemble)	✓ Observed (definitional)
<i>Non-independent consistency checks (same SX-dominated data; reported for transparency only)</i>					
Delay-μ correlation Pearson $r = 0.93$ (SX-leveraged), $n = 5$	Correlation test	Pearson $r = 0.932$ collapses to $r \approx 0.15$ when SX is excluded; Cook's distance > 1.0 for SX. Spearman $\rho = 0.3$ ($p = 0.31$) is not significant. Theil-Sen slope 86 ± 210 d overlaps zero. <i>Statistically inappropriate with $n = 5$ and one extreme leverage point. Not probative. Reported for transparency only.</i>	Not meaningful as formal test	✗ No (SX-driven; correlated with strand 1)	✓ Observed (reported for transparency)
Per-model α inference $\bar{\alpha}_{\text{lens}} \approx -0.055$	Parameter inference	Weighted mean α_{inferred} matches empirical coupling; scatter $\chi^2 = 0.66/7$ d.o.f. All 8 models ≤ 0 .	$p = 0.10$ vs. zero (one-sided)	✗ No (same ensemble)	✓ Observed
χ^2 model comparison GR vs. TEP ensemble fit	Goodness-of-fit	$\Delta\chi^2 = +1.6$ in favour of the proxy model; 46% wRMS reduction after proxy-model correction (5/8 models)	$p = 0.21$ (marginal)	✗ No (same ensemble)	✓ Observed
Directional-odds Bayes factor Sign data recast in odds form	Bayesian directional sign model	$\text{BF}_{10} = 31.9$ (all non-zero), 18.1 (blind-only), 10.5 (method-family-collapsed)	Odds support for one-sided sign excess	✗ No (same sign data)	✓ Observed (complementary to sign tests)
Correlated Significance Synthesis All tests structurally correlated	Synthesis Framework	Avoids double-dipping. Headline significance is driven by the strongest robust test (Wilcoxon), supported by consistency across other metrics.	$z = 2.2\sigma$ blind-only benchmark	—	✓ Methodological constraint
Loop SNR geometry Alpha-independent invariant	Structural prediction	SX loops: SNR = 63–66 at all $\alpha > 0$; geometric invariant of lens geometry	Geometric (no σ)	✓ Yes (theoretical)	✓ Structural
SN H0pe independent sensitivity Independent system	Predicted sensitivity	$\mathcal{R}_{\text{TEP}} \approx +1.90$ d, SNR = 2.12; requires independent 4th-image delay	$p \approx 0.017$ (predicted)	✓ Yes (different system)	✗ Not yet observed

None of these strands is individually decisive. The observed tests point in a coherent direction: the sign is right, the method-independence is right, and the implied coupling is self-consistent by construction. The Pearson correlation between delay and inverse-magnification ($r = 0.93$) is driven entirely by SX and is not statistically independent of the sign-based strand; it is reported for transparency but is not counted toward the headline significance. A complementary directional-odds analysis adds an odds-language view of the same sign data (directional BF_{10} from 10.5 to 31.9 across subsets), which improves interpretability but does not add a new independent dataset. Because several strands are correlated through the same SX-dominated sample, the most defensible headline remains the pre-specified primary non-parametric directional test: the blind-only Wilcoxon signed-rank test giving $p = 0.016$ ($\approx 2.2\sigma$), with the all-eight-model Wilcoxon ($p = 0.0078$, 2.4σ) reported as a supplementary consistency check.

Combining correlated metrics using standard meta-analysis (e.g., Fisher's method) is explicitly avoided to prevent artificial significance inflation.

The key probative point is that the direction of the SX residual matches the proxy-model prediction across seven modelling groups using five different codes, none of which had any knowledge of TEP when their predictions were made. The probability that independent random sign scatter would produce the blind-only pattern is $p = 0.016$ under the standard Wilcoxon test (all 6 non-zero blind residuals positive), with the supplementary all-eight-model test giving $p = 0.0078$. Shared lensing inputs and community-level modelling systematics prevent treating the models as fully independent draws; under a beta-binomial correlation model the break-even inter-model correlation for the blind-only Wilcoxon to reach $p = 0.05$ is $\rho \approx 0.08$ (step_11).

What this paper claims: a coherent, multi-pronged observational pattern — multiple observed evidence tests pointing in the direction predicted by the proxy model with no additional free parameters after calibration. This constitutes strong directional evidence, though not yet decisive model-selection evidence. This is the honest characterisation of the current evidence state.

What this paper does not claim: a detection of TEP. The lens model uncertainties of $\pm 16\text{--}60$ d prevent a $> 3\sigma$ conclusion from the current data. The existing data are *compatible with* the proxy model at the empirically determined coupling, and precision models targeting $\sigma < 5$ d would push this test to $> 5\sigma$.

4.8 Numerical Exercise: Low- H_0 Consistency Check

A cross-system numerical exercise applies the empirically determined proxy-model coupling ($\alpha_{\text{lens}} \approx -0.055$) to the low- H_0 measurements from lensed supernovae. Standard GR analyses of SN Refsdal (Kelly et al. 2023), SN Encore (Pierel et al. 2025), and SN H0pe (TD-only; Pierel et al. 2024) yield $H_0 \approx 61 - 67 \text{ km s}^{-1} \text{ Mpc}^{-1}$.

The proxy model predicts that systems where the most magnified image arrives first will have GR-inferred H_0 biased low, because the observed delay exceeds the geometric delay. Applying the empirically determined coupling, SN Refsdal shifts to $69.3 \text{ km s}^{-1} \text{ Mpc}^{-1}$, which would align with Planck if the proxy model were correct. **Important caveat:** this is not an independent confirmation because α_{lens} was empirically determined from the same SN Refsdal SX delay data (step_07). It is an internal consistency check: the proxy framework, calibrated on one feature of Refsdal, correctly predicts the H_0 shift from the same feature. SN Encore and SN H0pe receive only $\sim 0.1 \text{ km s}^{-1} \text{ Mpc}^{-1}$ shifts because their modest magnification contrasts and fewer independent delays suppress the per-delay gamma factors. The dominant proxy-model H_0 correction therefore comes from SN Refsdal. This differential prediction — strong correction for high-contrast multi-image systems, negligible correction for modest-contrast systems — is a falsifiable signature of the proxy model, but it is not independent evidence.

4.9 SN 2025wny: The Next Target

SN 2025wny ($z_s = 2.011$, $z_l = 0.375$, Johansson et al. 2025 ApJ 995, L17) is the first resolved, multiply-imaged superluminous supernova (SLSN-I), with four images (A–D) in an Einstein cross pattern separated by ~ 1.7 arcsec. With a magnification factor estimated at $\mu \sim 20\text{--}50$ for the brightest image, the system has a large potential contrast between images—precisely the regime where proxy-model predicted discrepancies are largest.

Unlike SN Refsdal, SN 2025wny does not yet have measured time delays. The discovery paper reports no time-resolved multi-image light curves. As a SLSN-I, its multi-month light curve evolution provides a natural clock for delay measurement from ground-based monitoring. A cross-to-arc algebraic loop (analogous to the S1–SX loop in SN Refsdal) predicts a proxy-model discrepancy of order 1–10 days at the measured coupling ($\alpha_{\text{lens}} \approx -0.055$), detectable with precision photometry once delays are measured to $\lesssim 1$ day precision.

4.10 Alternative Astrophysical Explanations

Before attributing the observed positive residual to the proxy model, it is necessary to consider whether conventional astrophysical effects could produce the same sign pattern across multiple independent modelling methods. Several alternative explanations are examined below.

4.10.1 Unmodelled Cluster Substructure

Unmodelled dark matter subhalos or line-of-sight mass concentrations could in principle introduce systematic errors in lens model predictions. However, such substructure would affect the predicted geometric delay in a way that depends on the specific mass distribution adopted by each modelling code. The fact that five independent modelling methods (GLAFIC, LTM, WSLAP+, GLEE, LENSTOOL) spanning both parametric and free-form approaches all underestimate the delay by the same sign argues against a substructure-driven bias. Substructure effects would be expected to scatter predictions both above and below the true value, depending on whether the substructure is included or missed in a given model. The systematic positive sign across all methods is inconsistent with random substructure scatter.

4.10.2 Line-of-Sight Mass Convergence

Mass structures along the line of sight to the cluster can introduce external convergence (κ_{ext}) that rescales all time delays by a common factor ($1 - \kappa_{\text{ext}}$). This is the external component of the Mass Sheet Degeneracy. Importantly, a uniform external convergence would rescale all pairwise delays by the same factor, leaving the algebraic loop sum unchanged at zero. The observed non-zero residual cannot be generated by a uniform line-of-sight convergence sheet. Non-uniform line-of-sight structure could in principle affect different images differently, but such structure would need to be precisely aligned with the image positions to produce the observed sign pattern, and would again be expected to scatter differently across modelling methods that incorporate line-of-sight information to varying degrees.

4.10.3 Microlensing-Induced Flux Bias

Microlensing by stars in cluster member galaxies can perturb observed fluxes, potentially biasing the flux-ratio-based magnification proxies used to infer Γ_t factors. This systematic is addressed in §2.4.1 and tested via the microlensing-nuisance Monte Carlo in §3.6.5. The key point is that microlensing perturbations at the tens-of-percent level can shift the inferred amplitude of $\Delta\Gamma$ but do not naturally invert the rank-ordering $\mu_{\text{SX}} < \mu_{\text{S4}}$ that sets the sign of the predicted SX residual. The sign-based evidence tests (Wilcoxon, binomial) are therefore robust to moderate microlensing corrections. Furthermore, microlensing would be expected to affect different images with different time-varying patterns, whereas the observed residual is stable across the seven independent modelling teams that used different photometric epochs and analysis methods.

4.10.4 Source-Size Effects

The finite size of the supernova host galaxy or the supernova light curve itself could in principle affect the inferred time delays if the lens models assume point-source approximations. However, Kelly et al. (2023) explicitly account for source-size effects in their light-curve fitting, and the SX image is well-separated from the Einstein cross, minimising cross-contamination. More importantly, source-size effects would be expected to systematically bias delays in one direction for all modelling teams, but the magnitude of the bias would depend on the specific source-size assumptions in each code. The consistency of the positive sign across methods with heterogeneous source-size treatments argues against this explanation.

4.10.5 Photometric Systematics

Systematic errors in the measured SN fluxes or in the model-predicted flux ratios could bias the magnification proxy estimates. However, the primary evidence test (§3.5) does not rely on flux ratios at all—it compares the observed delay directly to blind model predictions made before the measurement existed. The flux-ratio-based evidence (delay–magnification correlation, per-model alpha inference) is treated as supplementary and explicitly caveated. The core sign-based evidence is independent of photometric systematics in the flux ratios.

Taken together, these alternative explanations are either ruled out by the geometric structure of the blind-prediction residual test (uniform convergence cannot produce a differential residual), inconsistent with the multi-method sign pattern (substructure, line-of-sight), or robust to the specific test design (microlensing, photometric systematics). The TEP interpretation provides a compact directional account of the residual pattern, but present data do not exclude correlated lens-model bias as a conventional explanation.

4.10.6 Correlated Lens-Model Systematics

A referee's concern is that the blind lens models may not be statistically independent: they share common assumptions about the cluster mass distribution, the same HST imaging, and similar parametric halo prescriptions. If the models are correlated, the effective sample size is smaller than the nominal $N = 8$ (or $N = 7$ blind), and the significance of the sign tests drops.

This concern is quantified with a beta-binomial dependence model that treats the binary sign of each residual as an exchangeable outcome with intra-class correlation coefficient ρ . For the all-eight binomial sign test (7/8 positive, $p = 0.035$ under independence), the p-value rises above 0.05 once $\rho \gtrsim 0.03$. For the blind-only binomial (6/7 positive, $p = 0.063$ already), the nominal independent p-value is itself above the conventional threshold. The binomial sign test is therefore extremely fragile to even modest inter-model correlation.

The pre-specified primary test—the blind-only Wilcoxon signed-rank test—is more robust than the binomial sign test because it uses rank information, not merely signs. However, the Wilcoxon statistic does *not* have a closed-form variance under exchangeable intra-class correlation. The approximate scaling $n_{\text{eff}} = n/(1 + (n - 1)\rho)$, combined with $p \approx 1/2^{n_{\text{eff}}}$, is **mathematically unfounded** for the Wilcoxon distribution under dependence. The primary correlation-aware test is therefore the beta-binomial sign test reported above. A method-family block-bootstrap (Step 11), which resamples method families with replacement and computes the Wilcoxon statistic on each bootstrap draw, yields a distribution of dependence-adjusted p -values. This is the most defensible dependence-aware treatment of the rank data currently available.

The exact dependence-adjusted Wilcoxon p -value lies between the independent value ($p = 0.016$) and the beta-binomial sign-test value ($p = 0.063$ at $\rho = 0$), but its exact mapping to ρ requires a Monte Carlo permutation test respecting method-family clusters. Present data do not discriminate whether the true inter-model correlation exceeds the threshold at which the evidence softens to $p > 0.05$. Lens-modelling challenges for clusters such as H0LiCOW and TDCOSMO have demonstrated that different codes can disagree at the ~ 10 –20% level, suggesting some but not total independence. The TEP interpretation offers a single compact parameter ($\alpha_{\text{lens}} \approx -0.055$) that simultaneously explains the sign, magnitude, and directional odds of the SX residual, but a

conventional correlated-lens-model bias cannot be excluded without either (i) a larger ensemble of truly independent blind models, or (ii) a direct test of the symmetric-scatter assumption (e.g., comparing blind and post-blind predictions for the same system).

4.11 The Precision Roadmap to 5σ

The headline single-test significance from SN Refsdal is $z = 2.2\sigma$ (Wilcoxon signed-rank, the pre-specified primary non-parametric directional test, blind-only). The limiting factor is not the size of the proxy-model signal — the 14.5-day predicted shift is far above the 5.6-day measurement precision — but rather the large uncertainties in current GR lens models ($\sigma_{\text{model}} \approx 16\text{--}60$ d). Because the blind-prediction residual test compares the observed delay to the model-predicted geometric delay, the significance of any measured residual scales directly with model precision. Attempting to combine correlated p-values (e.g., via Fisher's method) is statistically invalid and would artificially inflate significance; the analysis therefore reports only the pre-specified primary non-parametric directional test as the benchmark.

Increased precision models are required to overcome this limitation. For cluster lenses, reaching $\sigma_{\text{model}} < 5$ d would decisively test the proxy-model prediction. To quantify the required precision, the ensemble significance was simulated for $N = 8$ independent models as a function of the average per-model uncertainty σ_{model} , assuming the true proxy-model signal is $\mathcal{R}_{\text{TEP/GR}} = 14.5$ d.

The roadmap shows that a clear detection threshold exists: if the community average uncertainty drops below $\sigma_{\text{model}} = 13.7$ d, the same 14.5-day mean residual crosses the 3σ "evidence" threshold. If models reach $\sigma_{\text{model}} = 8.2$ d, the exact same residual constitutes a 5σ "discovery" of potential-dependent temporal shear. At this precision, the binomial sign test becomes overwhelmingly decisive, as nearly all independent model predictions fall strictly below the observed delay.

5. Conclusion

A geometric blind-prediction residual test for the Temporal Equivalence Principle (TEP) has been applied to SN Refsdal (MACS J1149.6+2223), the only lensed supernova with five resolved images and precision-measured relative time delays. The key results are:

1. Under General Relativity, the algebraic loop sum is identically zero for all five independent image-triplet loops by construction. This identity holds for any theory with globally assignable arrival times per image, including the TEP ansatz. The genuine TEP observable is a blind-prediction residual — the discrepancy between observed delays and GR model predictions — not a closure violation.
2. The three loops constructed from the Einstein-cross images (S1–S4) predict residuals of 0.1–0.3 days at formal SNR $\approx 2\text{--}3$ under the log-magnification ansatz, using Kelly et al. (2023) flux ratios as magnification proxies and the measured coupling $\alpha_{\text{lens}} \approx -0.055$. These values are physically non-probative because the $\mu \rightarrow \kappa$ proxy fails in the inner cross (shear degeneracy renders magnification ordering uninformative about convergence).
3. The two loops incorporating image SX—which arrives 376 days after S1—yield predicted residuals of -8.5 days (S1–S2–SX, SNR = 66) and -14.5 days (S1–S4–SX, SNR = 63). The 376-day baseline amplifies the differential temporal shear between the most magnified cross image (S4) and the peripheral arc (SX) into an unambiguous signal far exceeding the 5.6-day measurement uncertainty.
4. The algebraic loop sum is immune to the Mass Sheet Degeneracy. A uniform convergence sheet rescales all delays by the same factor, leaving the loop sum unchanged at zero under both GR and TEP. The blind-prediction residual therefore constitutes an MSD-insensitive residual test, though still limited by GR lens-model precision and correlated modelling systematics.
5. The pre-specified primary non-parametric directional test is the blind-only Wilcoxon signed-rank test (all 6 non-zero residuals among the seven blind models are positive; $p = 0.016$, approximately 2.2σ), selected a priori because it equal-weights independent modelling groups and eliminates inverse-variance downweighting bias. The supplementary all-eight-model Wilcoxon (including one post-blind precision update) gives $p = 0.0078$ (2.4σ). The binomial sign test serves as a corroborating check (7/8 positive; $p = 0.035$). A complementary directional-odds analysis expresses the same sign pattern as one-sided Bayes factors ($\text{BF}_{10} = 31.9$ all non-zero; 18.1 blind-only; 10.5 method-family-collapsed), reinforcing directional support without introducing a new independent strand. Robustness checks preserve that directional pattern under model-dependence stress tests and 10%–30% microlensing-style flux perturbations, while a hierarchical Bayesian comparison with a proper GR-centred free-alpha prior gives non-decisive Bayes factors ($\text{BF} = 1.06$ baseline / 0.997 h0pe-informed for the fixed-alpha test of the specific SN Refsdal prediction; $\text{BF} = 0.615$ baseline / 0.492 h0pe-informed for the free-alpha model with a null-centred prior), indicating that present data remain in an inconclusive model-selection regime.
6. The measured TEP coupling ($\alpha_{\text{lens}} \approx -0.055$) provides a candidate resolution for the "Low H_0 " bias in SN Refsdal, shifting the inferred value from 66.6 to 69.3 $\text{km s}^{-1} \text{Mpc}^{-1}$ and bringing it into concordance with the Planck CMB standard. This is an internal consistency check, not an independent confirmation, because α_{lens} was calibrated on the same Refsdal data. SN Encore and SN H0pe receive only ~ 0.1 $\text{km s}^{-1} \text{Mpc}^{-1}$ corrections due to their smaller magnification contrasts and fewer independent delays. These are independent predictions (alpha calibrated on Refsdal, applied without refitting), though the shifts are negligible.
7. If independent delay measurements yield $|\mathcal{R}_{\text{obs}}(\text{S1}, \text{S4}, \text{SX})| < 1$ day, the linear log-magnification TEP ansatz is falsified in the strong-lensing domain. Conversely, a residual consistent with -14.5 days would constitute direct geometric evidence for potential-dependent temporal propagation once lens-model uncertainties are reduced sufficiently for decisive significance.
8. The newly discovered quadruply-imaged SLSN-I SN 2025wny ($z_s = 2.011$, Johansson et al. 2025) provides an analogous test target once time delays are measured. Given its magnification factor $\mu \sim 20\text{--}50$, predicted GR discrepancies are of order 1–15

days at the measured coupling ($\alpha_{\text{lens}} \approx -0.055$).

The blind-prediction residual test established here is a direct geometric probe of potential-dependent temporal propagation via the proxy model, with a clear falsification structure. The current evidence is suggestive but not yet decisive: directional consistency is strong, with the most robust single test yielding approximately 2.2σ significance ($p = 0.016$) from the blind-only Wilcoxon signed-rank test, while formal model selection is limited by lens-model uncertainty. Statistical power scales with both model precision and time-delay baseline; additional independent long-baseline multiply-imaged supernovae would be needed for a decisive test.

References

Strong Lensing Observations

- Kelly, P. L., Rodney, S. A., Treu, T., et al. 2015, *Science*, 347, 1123, "Multiple images of a highly magnified supernova formed by an early-type cluster galaxy lens"
- Kelly, P. L., Rodney, S., Treu, T., et al. 2023, *ApJ*, 948, 93, "The SN Refsdal Expansion Rate Measurement: A Powerful New Cosmic Ruler"
- Treu, T., Brammer, G., Diego, J. M., et al. 2016, *ApJ*, 817, 60, "The Refsdal Revelation: Predictions for the Return of the First Multiply Imaged Supernova"
- Grillo, C., Rosati, P., Suyu, S. H., et al. 2024, *ApJ*, 971, 49, "A High-precision Lens Model for MACS J1149.6+2223: Predicting the Reappearance of SN Refsdal"
- Frye, B. L., Pascale, M., Pierel, J., et al. 2024, *ApJ*, 961, 171, "The JWST Discovery of the Triply Imaged Type Ia "Supernova H0pe" and Observations of the Galaxy Cluster PLCK G165.7+67.0"
- Pierel, J. D. R., Frye, B. L., Pascale, M., et al. 2024, *ApJ*, 967, 50, "SN H0pe: The First Measurement of H0 from a Multiply Imaged Type Ia Supernova Discovered by JWST"
- Pierel, J. D. R., et al. 2025, (in prep), "SN Encore: A Lensed Supernova for Time Delay Cosmography"
- Grayling, M., Thorp, S., Mandel, K. S., et al. 2025, arXiv:2510.11719, "BayeSN-TD: Time Delay and H0 Estimation for Lensed SN H0pe"
- Johansson, J., Goobar, A., Suyu, S. H., et al. 2025, *ApJ*, 995, L17, "SN 2025wny: A Quadrupty Imaged Superluminous Supernova Discovered by JWST"
- Falco, E. E., Gorenstein, M. V., & Shapiro, I. I. 1985, *ApJ*, 289, L1, "Modeling of the Einstein Ring in MG 1131+0456"

Cosmology & H0 Measurements

- Planck Collaboration, Aghanim, N., Akrami, Y., et al. 2020, *A&A*, 641, A6, "Planck 2018 results. VI. Cosmological parameters"
- Riess, A. G., Yuan, W., Macri, L. M., et al. 2022, *ApJ*, 934, L7, "A Comprehensive Measurement of the Local Value of the Hubble Constant with 1 km/s/Mpc Uncertainty from the Hubble Space Telescope and the SH0ES Team"

Hubble Tension Reviews

- Di Valentino, E., Mena, O., Pan, S., et al. 2021, *Classical and Quantum Gravity*, 38, 153001, "In the realm of the Hubble tension—a review of solutions"

TEP Framework (This Series)

- Smawfield, M. L. 2025a, "Temporal Equivalence Principle: Dynamic Time & Emergent Light Speed" (Paper 1; in preparation)
- Smawfield, M. L. 2025e, "Temporal-Spatial Coupling in Gravitational Lensing" (Paper 5; in preparation)
- Smawfield, M. L. 2025g, "Universal Critical Density: Unifying Atomic, Galactic, and Compact Object Scales" (Paper 7; in preparation)
- Smawfield, M. L. 2025j, "What Do Precision Tests of General Relativity Actually Measure?" (Paper 10; in preparation)
- Smawfield, M. L. 2026a, "The Temporal Equivalence Principle: Suppressed Density Scaling in Globular Cluster Pulsars" (Paper 11; in preparation)

Appendix A: Theoretical Framework for Lensing Closures

A.1 Temporal Shear and the Lensing Potential

The Temporal Equivalence Principle (TEP) postulates that the rate of proper time flow for a photon traversing a gravitational potential Φ is scaled relative to the cosmological background rate by a factor Γ_t :

$$\Gamma_t(\Phi) = 1 + \alpha \frac{|\Phi|}{c^2} \quad (18)$$

where α is the lensing-sector effective coupling. The coupling is determined empirically from the SN Refsdal data: $\alpha_{\text{lens}} = -0.055 \pm 0.044$. In the strong lensing regime, the relevant potential is the projected gravitational potential $\psi(\boldsymbol{\theta})$ integrated along the line of sight. For singular isothermal sphere (SIS) or NFW profiles typical of cluster lenses, the projected potential depth scales logarithmically with the surface mass density Σ , which is directly related to the magnification μ .

No first-principles derivation of the log-magnification ansatz from the TEP scalar-field action through the lensing potential $\psi(\boldsymbol{\theta})$ currently exists. The mapping from projected convergence κ to observed magnification μ involves the full boundary-value problem of the scalar-field equation in a non-trivial cluster potential, including ellipticity, external shear, and substructure. Solving this transfer function is beyond the scope of the present work.

The following phenomenological scaling relation is therefore adopted as a strong-lensing-specific ansatz, analogous to the PPN strategy of treating the observable response separately from the microscopic coupling:

$$\Gamma_t(i) \approx 1 + \alpha \log_{10} \left(\frac{\mu_i}{\bar{\mu}} \right) \quad (19)$$

where μ_i is the absolute magnification of image i , and $\bar{\mu}$ is the mean magnification of the image system. This logarithmic scaling captures the essential feature that temporal shear is differential: it depends on the *ratio* of potential depths probed by different images. The test is a screen of this ansatz, not a measurement of the fundamental coupling.

A.2 Derivation of the Predicted GR Discrepancy

Let t_i^{GR} be the arrival time of image i predicted by General Relativity and t_0 the unlensed arrival time. Define the GR absolute delay $u_i = t_i^{\text{GR}} - t_0$ and the GR pairwise delay $\Delta t_{ij}^{\text{GR}} = t_j^{\text{GR}} - t_i^{\text{GR}} = u_j - u_i$. Under TEP, the observed arrival time is scaled by the path-specific shear Γ_i :

$$t_i^{\text{obs}} = t_0 + \Gamma_i(t_i^{\text{GR}} - t_0) = t_0 + \Gamma_i u_i \quad (20)$$

The observed pairwise delay between images i and j is therefore:

$$\Delta t_{ij}^{\text{obs}} = t_j^{\text{obs}} - t_i^{\text{obs}} = \Gamma_j u_j - \Gamma_i u_i \quad (21)$$

Because observed arrival times are physical clock readings, the sum of observed pairwise delays around any closed loop is identically zero:

$$\Delta t_{ij}^{\text{obs}} + \Delta t_{jk}^{\text{obs}} + \Delta t_{ki}^{\text{obs}} = (\Gamma_j u_j - \Gamma_i u_i) + (\Gamma_k u_k - \Gamma_j u_j) + (\Gamma_i u_i - \Gamma_k u_k) = 0 \quad (22)$$

The TEP signature is therefore not a failure of the observed delays to close, but a discrepancy between the observed delays and the delays predicted by GR lens models. The TEP predicted GR discrepancy is defined as the weighted sum of GR pairwise delays with coefficients $(\Gamma_m - 1)$:

$$\mathcal{R}_{\text{TEP}}(i, j, k) \equiv (\Gamma_i - 1) \Delta t_{ij}^{\text{GR}} + (\Gamma_j - 1) \Delta t_{jk}^{\text{GR}} + (\Gamma_k - 1) \Delta t_{ki}^{\text{GR}} \quad (23)$$

Substituting $\Delta t_{ij}^{\text{GR}} = u_j - u_i$ and expanding:

$$\mathcal{R}_{\text{TEP}} = (\Gamma_i - 1)(u_j - u_i) + (\Gamma_j - 1)(u_k - u_j) + (\Gamma_k - 1)(u_i - u_k) \quad (24)$$

Grouping terms by absolute delay:

$$\mathcal{R}_{\text{TEP}} = u_i(\Gamma_k - \Gamma_i) + u_j(\Gamma_i - \Gamma_j) + u_k(\Gamma_j - \Gamma_k) \quad (25)$$

This can also be written in the pairwise-delay form used in the main text by expanding $\Gamma = 1 + \delta\Gamma$ and using $\sum_{\text{loop}} \Delta t^{\text{GR}} = 0$:

$$\mathcal{R}_{\text{TEP}} = \sum_{\text{loop}} (1 + \delta\Gamma) \Delta t^{\text{GR}} = \sum_{\text{loop}} \Delta t^{\text{GR}} + \sum_{\text{loop}} \delta\Gamma \Delta t^{\text{GR}} = \sum_{\text{loop}} (\Gamma_m - 1) \Delta t_{mn}^{\text{GR}} \quad (26)$$

Explicitly for the triplet (i, j, k) :

$$\mathcal{R}_{\text{TEP}} = (\Gamma_i - 1)\Delta t_{ij} + (\Gamma_j - 1)\Delta t_{jk} + (\Gamma_k - 1)\Delta t_{ki} \quad (27)$$

where $\Delta t_{ij} \equiv \Delta t_{ij}^{\text{GR}}$. This equation demonstrates that the residual is generated purely by the *differences* in Γ around the loop. If the potential depth is constant ($\Gamma_i = \Gamma_j = \Gamma_k$), the residual vanishes. For the S1–S4–SX loop of SN Refsdal, where the other two delays are well-constrained by the Einstein-cross images, this weighted sum approximately equals the TEP-induced discrepancy in the SX–S1 delay.

A.3 Immunity to Mass Sheet Degeneracy

The Mass Sheet Degeneracy (MSD) corresponds to a transformation of the convergence $\kappa \rightarrow \lambda\kappa + (1 - \lambda)$, which rescales time delays by a factor λ :

$$\Delta t'_{ij} = \lambda \Delta t_{ij} \quad (28)$$

Substituting this into the predicted GR discrepancy definition:

$$\mathcal{R}' = \Delta t'_{ij} + \Delta t'_{jk} + \Delta t'_{ki} = \lambda(\Delta t_{ij} + \Delta t_{jk} + \Delta t_{ki}) \quad (29)$$

Since $\Delta t_{ij} + \Delta t_{jk} + \Delta t_{ki} = 0$ in GR, then $\mathcal{R}' = \lambda \cdot 0 = 0$.

Thus, the MSD cannot produce a non-zero loop sum. A measured non-zero blind-prediction residual $\mathcal{R}_{\text{obs}} \neq 0$ is therefore a robust signature of non-GR physics (specifically, potential-dependent temporal shear) that cannot be mimicked by standard lensing degeneracies.

Data Availability & Reproducibility

This work follows open-science practices. All results are fully reproducible from raw data using the documented pipeline. All numerical results, figures, and statistics are generated by deterministic Python scripts processing real observational data.

Repository & Code

GitHub Repository: github.com/matthewsmawfield/TEP-LENS

The repository contains a deterministic, version-controlled analysis pipeline with **21 analysis steps** (numbered 00–20) for geometric blind-prediction residual tests in multiply-imaged supernovae.

Repository Structure

```
TEP-LENS/
├── data/                # Raw light curves and lens models
│   └── sn_lensing/      # SN Refsdal and H0pe data
```

```
├── snh0pe/                                # SN H0pe light curves
├── scripts/
│   ├── steps/                            # Sequential analysis pipeline
│   │   ├── step_01_fetch_snh0pe_data.py
│   │   ├── step_02_gr_closure.py
│   │   ├── step_03_tep_closure.py
│   │   ├── step_04_plot_closure.py
│   │   ├── step_11_model_dependence.py
│   │   ├── step_13_bayes_model_comparison.py
│   │   └── run_all_steps.py              # Master pipeline runner
│   ├── simulations/                     # TEP lensing predictors
│   └── figures/                         # Publication-quality plotting
├── results/                             # Pipeline outputs and figures
├── site/
│   └── components/                      # HTML manuscript source
└── requirements.txt                     # Python dependencies
```

Data Provenance

Data Source	Provider	Access Method	Download Size	Reference
SN Refsdal Light Curves	Kelly et al. (2023)	Published data	~1 MB	ApJ 948, 93
SN H0pe Data	SN H0pe Collaboration	Public release	~500 KB	Via GitHub
TDCOSMO-2025 Chains	TDCOSMO Collaboration	Public chains	~50 MB (MCMC chains)	tdcosmo.github.io
H0LiCOW/TDCOSMO Legacy	H0LiCOW Collaboration	Public chains	~20 MB	Via repository

Total Download Size: ~75 MB for all primary data sources.
Data Provenance Log: Complete acquisition details maintained in `data/DATA_PROVENANCE.md`.

Reproduction Instructions

Quick Start (Full Reproduction)

```
# 1. Clone repository
git clone https://github.com/matthewsmawfield/TEP-LENS.git
cd TEP-LENS

# 2. Install dependencies
pip install -r requirements.txt

# 3. Run complete pipeline (21 steps, 00–20)
python scripts/steps/run_all_steps.py

# 4. Build manuscript
cd site
npm install
npm run build
```

System Requirements

Component	Minimum	Recommended	Tested On
CPU	4 cores	8+ cores	Apple M4 Pro (14-core)
RAM	8 GB	16 GB	24 GB (M4 Pro)
Storage	5 GB	10 GB	NVMe SSD
Runtime	~30-60 minutes		~30 minutes (M4 Pro)

Detailed Pipeline Steps

The analysis pipeline consists of **21 analysis steps** (numbered 00–20). Each step produces JSON outputs and logs for full traceability:

Data Acquisition & Closure Tests (Steps 00-04)

- **step_00_fetch_literature_and_cross_paper_data.py** — Fetch literature data and cross-paper reference values for TEP consistency checks
- **step_01_fetch_snh0pe_data.py** — Fetch SN H0pe light curve data from public release; parses multiply-imaged supernova magnitudes and time delays
- **step_02_gr_closure.py** — GR algebraic loop sum; computes the identically-zero loop baseline for GR with SN Refsdal and SN H0pe
- **step_03_tep_closure.py** — TEP predicted GR discrepancy; computes TEP-predicted GR discrepancies for each loop
- **step_04_plot_closure.py** — Generate predicted GR discrepancy plots (Figure 4 baseline vs discrepancy)

TDCOSMO & H0 Analysis (Steps 05-10)

- **step_05_tdcosmo_shear.py** — TDCOSMO temporal shear test; computes TEP-predicted delay shifts as a predicted-sensitivity consistency check
- **step_06_alpha_sensitivity.py** — α_0 parameter sensitivity analysis; tests how predicted GR discrepancies depend on TEP coupling
- **step_07_observed_vs_predicted.py** — Observed vs predicted time-delay comparison; quantifies TEP predictive accuracy
- **step_08_new_evidence.py** — Compile new evidence from independent lens systems
- **step_09_precision_roadmap.py** — Precision roadmap analysis; projects significance as a function of model precision and sample size
- **step_10_h0_tension.py** — H0 tension analysis in lensing context; compares TEP predictions to local vs CMB H0

Model Dependence & Robustness (Steps 11-15)

- **step_11_model_dependence.py** — Model dependence analysis; tests sensitivity to lens model assumptions
- **step_12_microlensing_robustness.py** — Microlensing robustness tests; quantifies microlensing contamination effects
- **step_13_bayes_model_comparison.py** — Bayesian model comparison; computes Bayes factors for GR vs TEP lens models
- **step_14_external_chain_ingestion.py** — External chain ingestion; imports TDCOSMO/H0LiCOW MCMC chains for joint analysis
- **step_15_external_informed_inflation.py** — External-informed inflation analysis; tests TEP effects on time-delay cosmography precision under external uncertainty priors

External Data & Completeness (Steps 16-20)

- **step_16_independence_tier_significance.py** — Independence tier significance; statistical tests for multiple independent lens systems
- **step_17_directional_odds.py** — Directional odds analysis; tests TEP predictions for specific image parity configurations
- **step_18_external_dataset_registry.py** — External dataset registry; catalogs all public lensed SN light curves
- **step_19_tdcosmo2025_ingestion.py** — TDCOSMO-2025 chain ingestion; imports latest time-delay cosmography constraints
- **step_20_external_completeness_synthesis.py** — External completeness synthesis; combines all lensing constraints with TEP

Each step produces JSON outputs with full metadata in `results/outputs/`, and execution logs are written to `logs/` with timestamps for complete traceability. Run all steps via: `python scripts/steps/run_all_steps.py`

Key Analysis Outputs

- `results/outputs/step_02_gr_closure.json` — GR algebraic loop sums
- `results/outputs/step_03_tep_closure.json` — TEP predicted GR discrepancies
- `results/outputs/step_13_bayes_model_comparison.json` — Bayesian comparison

Software Versions

- **Python** 3.10+
- **NumPy** 1.24+
- **SciPy** 1.10+
- **Pandas** 2.0+
- **Matplotlib** 3.7+